

PC1 (2024/2025) - ALGEBRA I

EXERCISE SET 04 — SOLUTIONS

Exercise 1. On $\mathbb{R}$, we define the binary operation $*$ by:

$$\forall x, y \in \mathbb{R}, \quad x * y = x + y + \sin(xy).$$

1. Is this operation commutative?
2. Does it have a neutral element?
3. Let $f(x) = x + 3 + \sin(3x)$. Compute $f(-4)$, $f(-3.8)$ and $f(-3.1)$.
4. What can we deduce from this?
5. Deduce that $*$ is not associative.

Solution:

1. Let $x, y \in \mathbb{R}$. Since $xy = yx$ and $\sin(xy) = \sin(yx)$, we have

$$x * y = x + y + \sin(xy) = y + x + \sin(yx) = y * x.$$

Thus $*$ is commutative.

2. Let $e \in \mathbb{R}$, for all $x \in \mathbb{R}$,

$$x * e = x \iff x + e + \sin(xe) = x,$$

which gives $e + \sin(xe) = 0$ for all x . For $e = 0$, this holds, so 0 is a neutral element. The uniqueness of the neutral element was established in the lecture.

3. We have

$$f(-4) \approx -0.4634, \quad f(-3.8) \approx 0.1193, \quad f(-3.1) \approx -0.2245.$$

4. The function f is continuous on $\mathbb{R}$, hence on $[-4, -3.8]$ and $[-3.8, -3.1]$. On $[-4, -3.8]$: $f(-4) < 0$ and $f(-3.8) > 0$, so by the Intermediate Value Theorem there exists $a \in (-4, -3.8)$ with $f(a) = 0$. On $[-3.8, -3.1]$: $f(-3.8) > 0$ and $f(-3.1) < 0$, so there exists $b \in (-3.8, -3.1)$ with $f(b) = 0$. We have $a \neq b$ and $a * 3 = b * 3 = 0$.

5. Using a and b from above with $a \neq b$ and $a * 3 = b * 3 = 0$:

$$(a * 3) * b = 0 * b = b,$$

$$a * (3 * b) = a * 0 = a.$$

Since $a \neq b$, we have $(a * 3) * b \neq a * (3 * b)$. Therefore $*$ is not associative.

Exercise 2.

1. Let $\mathcal{H}$ be the set of binary relations on a nonempty set E . For $R, S \in \mathcal{H}$ define $T = R * S$ by

$$\forall x, y \in E, \quad xTy \iff \exists z \in E (xRz \wedge zSy).$$

2. On $\mathbb{Z}$ define

$$x * y = xy + \lfloor \varphi x \rfloor \lfloor \varphi y \rfloor \quad (x, y \in \mathbb{Z}),$$

where $\varphi = \frac{1 + \sqrt{5}}{2}$ and $\lfloor \cdot \rfloor$ is the floor function.

Solution:

1. Let $R, S, U \in \mathcal{H}$ and fix $x, y \in E$. Then

$$\begin{aligned} x((R * S) * U)y &\iff \exists w \in E (x(R * S)w \wedge wUy) \\ &\iff \exists w \in E (\exists z \in E (xRz \wedge zSw) \wedge wUy) \iff \exists z \in E (xRz \wedge \exists w \in E (zSw \wedge wUy)) \\ &\iff \exists z \in E (xRz \wedge z(S * U)y) \iff x(R * (S * U))y. \end{aligned}$$

Since this holds for all $x, y \in E$ we conclude $(R * S) * U = R * (S * U)$. Thus $*$ is associative on $\mathcal{H}$.

2. Define, for $n \in \mathbb{Z}$,

$$A(n) := \lfloor \varphi n \rfloor, \quad r(n) := \varphi n - A(n) \in [0, 1).$$

For $x, y, z \in \mathbb{Z}$ set

$$S_{xy} := xy + A(x)A(y), \quad Q_{xy} := xA(y) + yA(x) + A(x)A(y).$$

We first prove the key identity

$$A(S_{xy}) = Q_{xy}. \quad (*)$$

Compute

$$\varphi S_{xy} - Q_{xy} = \varphi(xy + A(x)A(y)) - (xA(y) + yA(x) + A(x)A(y)).$$

Substituting $A(t) = \varphi t - r(t)$ and using $\varphi^2 = \varphi + 1$, a straightforward expansion and cancellation gives

$$\varphi S_{xy} - Q_{xy} = r(x)r(y)(\varphi - 1).$$

Since $0 \leq r(x), r(y) < 1$ and $\varphi - 1 = \frac{1}{\varphi} \in (0, 1)$, it follows that

$$0 \leq \varphi S_{xy} - Q_{xy} < 1,$$

whence $\lfloor \varphi S_{xy} \rfloor = Q_{xy}$ and therefore $(*)$ holds.

Now compute both parenthesizations:

$$(x * y) * z = (xy + A(x)A(y))z + A(xy + A(x)A(y))A(z) = xyz + A(x)A(y)z + Q_{xy}A(z),$$

$$x * (y * z) = x(yz + A(y)A(z)) + A(x)A(yz + A(y)A(z)) = xyz + xA(y)A(z) + A(x)Q_{yz}.$$

Using the definitions of Q_{xy} and Q_{yz} , a direct expansion (term-by-term rearrangement) shows

$$A(x)A(y)z + Q_{xy}A(z) = xA(y)A(z) + A(x)Q_{yz},$$

hence $(x * y) * z = x * (y * z)$ for all $x, y, z \in \mathbb{Z}$. Therefore $*$ is associative on $\mathbb{Z}$.

Exercise 3.

1. Show that $(G, *)$ is a group in each of the following cases:

(a) $G = \mathbb{R}^2 \setminus \{(0, 0)\}$ with

$$(a, b) * (c, d) = (ac - bd, bc + ad).$$

(b) $G =] - 1, 1[$ with

$$x * y = \frac{x + y}{1 + xy}.$$

2. Let E be nonempty. Show that $(\mathcal{P}(E), \Delta)$, where

$$A \Delta B = (A \cup B) \setminus (A \cap B) \quad (A, B \subseteq E),$$

is an abelian group.

Closure. Let $(a, b), (c, d) \in G$. Then $a, b, c, d \in \mathbb{R}$ and $(a, b) \neq (0, 0)$, $(c, d) \neq (0, 0)$. We have

$$(a, b) * (c, d) = (ac - bd, bc + ad).$$

If $(ac - bd, bc + ad) = (0, 0)$, then $ac - bd = 0$ and $bc + ad = 0$. Multiplying the first equation by a and the second by b and adding gives

$$a^2c - abd + b^2c + abd = (a^2 + b^2)c = 0.$$

Since $(a, b) \neq (0, 0)$, we have $a^2 + b^2 > 0$, hence $c = 0$. Substituting $c = 0$ into $ac - bd = 0$ yields $-bd = 0$, so $b = 0$ or $d = 0$. If $b = 0$ then $(a, b) = (a, 0)$ with $a \neq 0$, and $bc + ad = 0$ gives $ad = 0 \Rightarrow d = 0$, hence $(c, d) = (0, 0)$, contradiction. If $d = 0$ then $(c, d) = (c, 0)$ with $c = 0$ from above, so $(c, d) = (0, 0)$, contradiction. Therefore $(ac - bd, bc + ad) \neq (0, 0)$ and belongs to G .

Associativity. Let $(a, b), (c, d), (e, f) \in G$. First compute

$$(a, b) * (c, d) = (ac - bd, bc + ad).$$

Then

$$((a, b) * (c, d)) * (e, f) = ((ac - bd)e - (bc + ad)f, (bc + ad)e + (ac - bd)f).$$

On the other hand,

$$(c, d) * (e, f) = (ce - df, de + cf),$$

and

$$(a, b) * ((c, d) * (e, f)) = (a(ce - df) - b(de + cf), b(ce - df) + a(de + cf)).$$

Expanding both results gives the same pair:

$$(ace - bde - adf - bcf, bce - bdf + ade + acf),$$

hence

$$((a, b) * (c, d)) * (e, f) = (a, b) * ((c, d) * (e, f)).$$

Thus $*$ is associative.

Identity. Seek $(u, v) \in G$ such that $(a, b) * (u, v) = (a, b)$ for every $(a, b) \in G$. This gives the system

$$au - bv = a, \quad bu + av = b \quad (\forall a, b).$$

Evaluate the system at $(a, b) = (1, 0)$: $u = 1$, $v = 0$. Hence $(1, 0)$ is the neutral element; directly check $(a, b) * (1, 0) = (a, b)$.

Inverse. Given fixed $(a, b) \in G$, determine (u, v) with

$$(a, b) * (u, v) = (1, 0)$$

$$(au - bv, bu + av) = (1, 0)$$

i.e.

$$au - bv = 1, \quad bu + av = 0.$$

Multiply the first equation by a and the second by b and add:

$$(a^2 + b^2)u = a,$$

so

$$u = \frac{a}{a^2 + b^2}.$$

Multiply the first equation by b and the second by a and subtract the second from the first:

$$-(a^2 + b^2)v = b,$$

hence

$$v = -\frac{b}{a^2 + b^2}.$$

Thus every $(a, b) \in G$ has the inverse $(\frac{a}{a^2 + b^2}, -\frac{b}{a^2 + b^2}) \in G$.

Commutativity. Follows from commutativity of complex multiplication.

Therefore $(G, *)$ is a group.

1.(b) $G =]-1, 1[$, $x * y = \frac{x + y}{1 + xy}$.

Closure. For $x, y \in (-1, 1)$ we have $xy > -1$, so $1 + xy > 0$ and the fraction is defined. Moreover

$$1 - \frac{x + y}{1 + xy} = \frac{(1 - x)(1 - y)}{1 + xy} > 0, \quad 1 + \frac{x + y}{1 + xy} = \frac{(1 + x)(1 + y)}{1 + xy} > 0,$$

hence $-1 < \frac{x + y}{1 + xy} < 1$ and $x * y \in (-1, 1)$.

Associativity. Compute

$$(x * y) * z = \frac{\frac{x + y}{1 + xy} + z}{1 + \frac{x + y}{1 + xy} z} = \frac{x + y + z + xyz}{1 + xy + yz + zx},$$

and

$$x * (y * z) = \frac{x + \frac{y + z}{1 + yz}}{1 + x \frac{y + z}{1 + yz}} = \frac{x + y + z + xyz}{1 + xy + yz + zx}.$$

Both parenthesizations yield the same expression, so associativity holds.

Identity. Seek $e \in (-1, 1)$ with $(x * e) = x$ for all $x \in (-1, 1)$. Solve

$$\frac{x + e}{1 + xe} = x \implies x + e = x + x^2 e \implies e(1 - x^2) = 0 \quad (\forall x).$$

Taking $x = 0$ gives $e = 0$. Thus 0 is the neutral element.

Inverse. For fixed $x \in (-1, 1)$ solve $x * y = 0$:

$$\frac{x + y}{1 + xy} = 0 \implies x + y = 0 \implies y = -x,$$

and $-x \in (-1, 1)$. Therefore the inverse of x is $-x$.

Commutativity. Immediate from the symmetry of the formula.

Therefore $(G, *)$ is a group.

2. $(\mathcal{P}(E), \Delta)$, with $A \Delta B = (A \cup B) \setminus (A \cap B)$.

Closure. If $A, B \subseteq E$ then $(A \cup B) \setminus (A \cap B) \subseteq (A \cup B) \subseteq E$, so $A \Delta B \in \mathcal{P}(E)$.

Associativity. Let $A, B, C \subseteq E$. We prove

$$(A \Delta B) \Delta C = A \Delta (B \Delta C)$$

by two inclusions.

First, let $x \in (A \Delta B) \Delta C$. By definition exactly one of the following holds:

(1) $x \in A\Delta B$ and $x \notin C$, or

(2) $x \notin A\Delta B$ and $x \in C$.

If (1) holds then $x \in (A \cup B)$ and $x \notin (A \cap B)$, so either (a) $x \in A$ and $x \notin B$, or (b) $x \in B$ and $x \notin A$. In case (a), since $x \notin C$ we have $x \in A$ and $x \notin (B \cup C)$ or $x \notin (B \cap C)$ as required; hence $x \in A\Delta(B\Delta C)$. In case (b), since $x \notin C$ we get $x \in B$ and $x \notin C$, so $x \in B\Delta C$ and $x \notin A$, hence $x \in A\Delta(B\Delta C)$. If (2) holds then either (i) $x \in A \cap B$ or (ii) $x \notin A \cup B$. In subcase (i) we have $x \in A$ and $x \in B$ and $x \in C$, so $x \notin B\Delta C$ and $x \in A$, hence $x \in A\Delta(B\Delta C)$. In subcase (ii) we have $x \notin A$, $x \notin B$, and $x \in C$, so $x \in B\Delta C$ and $x \notin A$, hence $x \in A\Delta(B\Delta C)$. Thus every possibility yields $x \in A\Delta(B\Delta C)$.

Second, let $x \in A\Delta(B\Delta C)$. Then exactly one of the following holds:

(3) $x \in A$ and $x \notin B\Delta C$, or

(4) $x \notin A$ and $x \in B\Delta C$.

If (3) holds then $x \in A$ and $x \notin (B \cup C)$ or $x \in B \cap C$. If $x \notin (B \cup C)$ then $x \in A$ and $x \notin B$, $x \notin C$, so $x \in A\Delta B$ and $x \notin C$, hence $x \in (A\Delta B)\Delta C$. If $x \in B \cap C$ then $x \in A$, $x \in B$, and $x \in C$, so $x \notin A\Delta B$ and $x \in C$, hence $x \in (A\Delta B)\Delta C$. If (4) holds then $x \in B\Delta C$ and $x \notin A$. If $x \in B\Delta C$ then either (i) $x \in B$ and $x \notin C$ (so $x \in A\Delta B$ and $x \notin C$), or (ii) $x \in C$ and $x \notin B$ (so $x \notin A\Delta B$ and $x \in C$). In both subcases $x \in (A\Delta B)\Delta C$. Therefore $A\Delta(B\Delta C) \subseteq (A\Delta B)\Delta C$.

Combining both inclusions yields $(A\Delta B)\Delta C = A\Delta(B\Delta C)$.

Identity. Find $E_0 \subseteq E$ with $A\Delta E_0 = A$ for all $A \subseteq E$. Take $A = \emptyset$: $\emptyset\Delta E_0 = E_0$ must equal $\emptyset$, hence $E_0 = \emptyset$. Check: for any A ,

$$A\Delta\emptyset = (A \cup \emptyset) \setminus (A \cap \emptyset) = A \setminus \emptyset = A.$$

Inverse. For given $A \subseteq E$ determine $B \subseteq E$ with $A\Delta B = \emptyset$. By definition

$$(A \cup B) \setminus (A \cap B) = \emptyset \iff A \cup B \subseteq A \cap B.$$

Thus $A \subseteq B$ and $B \subseteq A$, so $A = B$. Therefore each A is its own inverse.

Commutativity. Immediate from the symmetric definition $A\Delta B = (A \cup B) \setminus (A \cap B)$.

Therefore $(\mathcal{P}(E), \Delta)$ is an abelian group.

Exercise 4.

1. We consider the set $S = \{f_1, f_2, f_3, f_4, f_5, f_6\}$ of maps from $\mathbb{R} \setminus \{0, 1\}$ into $\mathbb{R} \setminus \{0, 1\}$, where

$$f_1(x) = x, \quad f_2(x) = 1 - x, \quad f_3(x) = \frac{1}{x}, \quad f_4(x) = \frac{1}{1 - x}, \quad f_5(x) = \frac{x - 1}{x}, \quad f_6(x) = \frac{x}{x - 1}.$$

(a) Write the multiplication table of S (composition of maps) and show that $(S, \circ)$ is a group.

(b) What are the subgroups of S ?

2. Write the multiplication table of a group of order 2 and a group of order 3.

Solution.

1.(a) Composition table for S (rows: left factor; columns: right factor):

$\circ$	f_1	f_2	f_3	f_4	f_5	f_6
f_1	f_1	f_2	f_3	f_4	f_5	f_6
f_2	f_2	f_1	f_5	f_6	f_3	f_4
f_3	f_3	f_4	f_1	f_2	f_6	f_5
f_4	f_4	f_3	f_6	f_5	f_1	f_2
f_5	f_5	f_6	f_2	f_1	f_4	f_3
f_6	f_6	f_5	f_4	f_3	f_2	f_1

Verification that $(S, \circ)$ is a group.

Closure. Every composition $f_i \circ f_j$ equals one of $f_1, \dots, f_6$ (see the table), so S is closed under composition.

Associativity. Composition of maps is associative, hence the operation $\circ$ is associative on S .

Identity. $f_1(x) = x$ is the identity: for every $f \in S$, $f_1 \circ f = f \circ f_1 = f$.

Inverses. From the table (entries equal to f_1 on the diagonal of the appropriate row/column) one reads the inverses:

$$f_1^{-1} = f_1, \quad f_2^{-1} = f_2, \quad f_3^{-1} = f_3, \quad f_4^{-1} = f_5, \quad f_5^{-1} = f_4, \quad f_6^{-1} = f_6.$$

Thus every element of S has an inverse in S .

Therefore $(S, \circ)$ is a group.

1.(b) Subgroups of S .

The *order* of an element $f \in S$ is the smallest positive integer n such that $f^n = \underbrace{f \circ f \circ \dots \circ f}_{n \text{ times}} = f_1$ (the identity map).

From the table or by direct computation:

$$\begin{aligned} \text{ord}(f_1) &= 1 && \text{(identity)}, \\ \text{ord}(f_2) &= 2 && \text{since } f_2 \circ f_2 = f_1, \\ \text{ord}(f_3) &= 2 && \text{since } f_3 \circ f_3 = f_1, \\ \text{ord}(f_6) &= 2 && \text{since } f_6 \circ f_6 = f_1, \\ \text{ord}(f_4) &= 3 && \text{since } f_4 \circ f_4 = f_5, f_4 \circ f_5 = f_1, \\ \text{ord}(f_5) &= 3 && \text{since } f_5 \circ f_5 = f_4, f_5 \circ f_4 = f_1. \end{aligned}$$

A subgroup must contain the identity and be closed under composition and inverses. From the element orders:

- Elements of order 2 generate subgroups of order 2: $\{f_1, f_2\}$, $\{f_1, f_3\}$, $\{f_1, f_6\}$. - Elements of order 3 generate subgroups of order 3: $\{f_1, f_4, f_5\}$ (generated by either f_4 or f_5). - The trivial subgroup $\{f_1\}$ and the whole group S are always subgroups.

Thus all subgroups are:

$$\{f_1\}, \{f_1, f_2\}, \{f_1, f_3\}, \{f_1, f_6\}, \{f_1, f_4, f_5\}, S.$$

2. Multiplication tables for a group of order 2 and a group of order 3.

Group of order 2 (write elements e, a with $a^2 = e$):

	e	a
e	e	a
a	a	e

Group of order 3 (cyclic, write elements e, b, b^2 with $b^3 = e$):

	e	b	b^2
e	e	b	b^2
b	b	b^2	e
b^2	b^2	e	b

Exercise 5

Let $(G, *)$ be a set equipped with a binary operation $*$ satisfying:

$$\forall a \in G, \exists e \in G \text{ such that } a * e = a \quad (\text{right neutral}),$$

$$\forall a \in G, \exists a^{-1} \in G \text{ such that } a * a^{-1} = e \quad (\text{right inverse}).$$

Show that $(G, *)$ is a group.

First, we show that the right neutral element is unique. Indeed, suppose $e_1 \in G$ is such that

$$\forall a \in G, \quad a * e_1 = a.$$

Then, taking $a = e$ (where e is the neutral element), we have:

$$e * e_1 = e.$$

Multiplying on the right by e_1 :

$$e_1 * e * e_1 = e_1 * e = e_1.$$

Therefore:

$$e_1^2 = e_1.$$

But then:

$$\begin{aligned} e_1 &= e_1 * e = e_1 * (e_1 * e_1^{-1}) = (e_1 * e_1) * e_1^{-1} \\ &\Leftrightarrow e_1 = e_1^2 * e_1^{-1} = e_1 * e_1^{-1} = e. \end{aligned}$$

So the neutral element is unique.

Next, we show that a right inverse is also a left inverse. Let $a \in G$. Then:

$$(a^{-1} * a) * (a^{-1} * a) = a^{-1} * (a * a^{-1}) * a = a^{-1} * a.$$

By the uniqueness of the neutral element, we have:

$$a^{-1} * a = e.$$

Thus the right inverse is a left inverse.

The element e is also a left neutral element. Indeed, for all $a \in G$:

$$e * a = (a * a^{-1}) * a = a * (a^{-1} * a) = a * e = a.$$

Finally, we show that the inverse is unique. For $a \in G$, let $y \in G$ such that $a * y = e$. Then:

$$a * y = a * a^{-1}.$$

Multiplying on the left by a^{-1} :

$$a^{-1} * (a * y) = a^{-1} * (a * a^{-1}) \Rightarrow (a^{-1} * a) * y = (a^{-1} * a) * a^{-1}.$$

Since $a^{-1} * a = e$, we obtain:

$$e * y = e * a^{-1} \Rightarrow y = a^{-1}.$$

Therefore, the inverse of a is unique.

We conclude that $(G, *)$ is a group.

Exercise 6. Let $(G, \cdot)$ be a group and e its neutral element.

1. Show that for any $a \in G$ and $n, m \in \mathbb{Z}$ we have $a^n a^m = a^{n+m}$.

2. Show that if every element of G is its own inverse, then G is abelian.
3. Assume that G has even order. Prove that there exists $a \in G$, $a \neq e$, such that $a^2 = e$.
4. Assume that G is finite. Show that there exists a positive integer m such that $a^m = e$ for all $a \in G$.
5. Show that if for all $a, b \in G$ we have $(ab)^2 = a^2b^2$, then G is abelian.

Solution.

1. We first recall the standard definition of integer powers in a group:

$$a^0 = e, \quad a^n = \underbrace{a \cdot a \cdots a}_{n \text{ factors}} \quad (n > 0), \quad a^{-n} = (a^n)^{-1} \quad (n > 0).$$

We prove $a^n a^m = a^{n+m}$ for all $n, m \in \mathbb{Z}$.

Case 1: $n, m \geq 0$. This is immediate from the definition of positive powers (concatenation of factors).

Case 2: $n \geq 0, m < 0$. Write $m = -k$ with $k > 0$. Then

$$a^n a^m = a^n a^{-k} = a^n (a^k)^{-1}.$$

If $n \geq k$ then $a^n = (a^{n-k})a^k$ so $a^n (a^k)^{-1} = a^{n-k} = a^{n+(-k)}$. If $n < k$ then

$$a^n (a^k)^{-1} = ((a^{k-n})^{-1}) = a^{-(k-n)} = a^{n+(-k)}.$$

Thus $a^n a^m = a^{n+m}$.

Case 3: $n, m < 0$. Put $n = -r, m = -s$ with $r, s > 0$. Then

$$a^n a^m = a^{-r} a^{-s} = (a^r)^{-1} (a^s)^{-1} = (a^s a^r)^{-1} = a^{-(r+s)} = a^{n+m},$$

using the already proven positive case on $a^s a^r$ and the rule $(xy)^{-1} = y^{-1}x^{-1}$.

Combining cases yields $a^n a^m = a^{n+m}$ for all integers n, m .

2. Assume $x^{-1} = x$ for every $x \in G$ (equivalently $x^2 = e$ for all x). For arbitrary $a, b \in G$ we have

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba,$$

where we used $(ab)^{-1} = b^{-1}a^{-1}$ and the hypothesis $a^{-1} = a, b^{-1} = b$. Hence G is abelian.

3. Suppose $|G|$ is even. Consider the map $\iota : G \rightarrow G$ given by $\iota(x) = x^{-1}$. This map is a bijection (inverse is itself). The fixed points of ι are exactly the elements satisfying $x = x^{-1}$, i.e. those with $x^2 = e$ (these include e). The remaining elements come in pairs $\{x, x^{-1}\}$ with $x \neq x^{-1}$, so the number of non-fixed elements is even. If the only fixed element were e , then $|G| = 1 + (\text{even})$ would be odd, contradicting that $|G|$ is even. Therefore there exists a fixed point $a \neq e$ with $a = a^{-1}$, hence $a^2 = e$.

4. Assume G is finite. For each $a \in G$ the (finite) subgroup $\langle a \rangle$ has finite order, say $\text{ord}(a)$. Consider the finite set of positive integers $\{\text{ord}(a) \mid a \in G\}$. Let

$$m = \text{lcm}\{\text{ord}(a) : a \in G\},$$

the least common multiple of all element orders (well-defined since the set is finite). For each $a \in G$ we have $\text{ord}(a) \mid m$, hence $a^m = e$. Thus the chosen m satisfies $a^m = e$ for all $a \in G$.

(As an alternative one may take $m = |G|!$ or $m = \prod_{a \in G} \text{ord}(a)$; these also work.)

5. Assume $(ab)^2 = a^2b^2$ for all $a, b \in G$. Expand the left-hand side:

$$(ab)^2 = abab.$$

Hence for all a, b ,

$$abab = a^2b^2 = aabb.$$

Multiply on the left by a^{-1} and on the right by b^{-1} :

$$a^{-1}(abab)b^{-1} = a^{-1}(aabb)b^{-1}.$$

The left side simplifies to ba , and the right side simplifies to ab . Therefore $ba = ab$ for all $a, b \in G$, so G is abelian.

Exercise 7. Let $(G, \cdot)$ be a group.

1. Show that for all $a, b \in G$, the elements ab and ba have the same order.
2. Suppose $(G, \cdot)$ is finite and abelian. Let $x, y \in G$ with $|x| = n$ and $|y| = m$. Show that if $\gcd(n, m) = 1$, then $|xy| = mn$.
3. Assume that an element $x \in G$ has $|x| = rs$. Find $|x^r|$.
4. If $|x| = n$, what is $|x^r|$ in general?

Solution.

1. Let $a, b \in G$ and set $|ab| = k$. Then $(ab)^k = e$. Multiply on the left by b and on the right by b^{-1} to obtain

$$b(ab)^kb^{-1} = (ba)^k = beb^{-1} = e.$$

By the definition of order (minimal positive exponent giving e), this shows $|ba| \mid k$.

Conversely, if $|ba| = m$ then $(ba)^m = e$, and the same manipulation gives $(ab)^m = e$, hence $|ab| \mid m$. Therefore $m \mid k$ and $k \mid m$, so $k = m$. Thus $|ab| = |ba|$.

2. Since G is abelian, $(xy)^t = x^t y^t$ for every integer t . In particular

$$(xy)^{nm} = x^{nm} y^{nm} = e \cdot e = e,$$

so $|xy| \mid nm$.

Let $d := |xy|$. Then $(xy)^d = e$, hence $x^d y^d = e$ and therefore $x^d = y^{-d}$. Thus $x^d \in \langle x \rangle \cap \langle y \rangle$. If $z \in \langle x \rangle \cap \langle y \rangle$ then $|z|$ divides both n and m , so $|z| \mid \gcd(n, m) = 1$, hence $z = e$. Therefore the intersection $\langle x \rangle \cap \langle y \rangle$ is $\{e\}$, and from $x^d = y^{-d}$ we deduce $x^d = e$ and $y^d = e$.

Consequently $n \mid d$ and $m \mid d$. Since $\gcd(n, m) = 1$ this implies $nm \mid d$. Combining with $d \mid nm$ yields $d = nm$. Hence $|xy| = mn$.

3. Let $|x| = rs$ and set $m := |x^r|$. First

$$(x^r)^s = x^{rs} = e,$$

so $m \mid s$. On the other hand $(x^r)^m = e$ implies $x^{rm} = e$, so $rs \mid rm$, whence $s \mid m$. Therefore $m \mid s$ and $s \mid m$, so $m = s$. Thus $|x^r| = s$.

4. Let $|x| = n$ and $r \in \mathbb{Z}$. Put $d = \gcd(n, r)$ and write $n = dn'$, $r = dr'$ so $\gcd(n', r') = 1$. We claim

$$|x^r| = \frac{n}{\gcd(n, r)} = \frac{n}{d} = n'.$$

Indeed, using $n = dn'$ we get

$$(x^r)^{n'} = x^{rn'} = x^{dr'n'} = x^{r'(dn')} = x^{r'n} = (x^n)^{r'} = e,$$

so $|x^r| \mid n'$. Conversely, if $|x^r| = m$ then $(x^r)^m = e$, i.e. $x^{rm} = e$, so $n \mid rm$. Writing $n = dn'$, $r = dr'$ this becomes $dn' \mid dr'm$, hence $n' \mid r'm$. Since $\gcd(n', r') = 1$ we conclude $n' \mid m$. Thus $n' \mid m$ and $m \mid n'$, so $m = n'$. This proves the formula

$$|x^r| = \frac{n}{\gcd(n, r)}.$$

Exercise 8. We define on $\mathbb{R}^2$ the binary operation $\oplus$ by

$$\forall (x, y), (x', y') \in \mathbb{R}^2, \quad (x, y) \oplus (x', y') = (x + x', y + y' + 2xx').$$

1. Show that $(\mathbb{R}^2, \oplus)$ is an abelian group.
2. Show that $H = \{(x, x^2) \mid x \in \mathbb{R}\}$ is a subgroup of $(\mathbb{R}^2, \oplus)$.
3. Show that the map $\Phi : (\mathbb{R}, +) \rightarrow (H, \oplus)$ defined by $\Phi(x) = (x, x^2)$ is an isomorphism of groups.

Solution.

1. *Closure.* Let $(x, y), (x', y') \in \mathbb{R}^2$. By definition

$$(x, y) \oplus (x', y') = (x + x', y + y' + 2xx') \in \mathbb{R}^2,$$

so $\oplus$ is closed on $\mathbb{R}^2$.

Associativity. Compute both parenthesizations and compare coordinates.

First

$$(x, y) \oplus (x', y') = (x + x', y + y' + 2xx'),$$

hence

$$((x, y) \oplus (x', y')) \oplus (x'', y'') = ((x + x') + x'', (y + y' + 2xx') + y'' + 2(x + x')x'').$$

The second coordinate expands to

$$y + y' + 2xx' + y'' + 2(x + x')x'' = y + y' + y'' + 2xx' + 2xx'' + 2x'x''.$$

On the other hand

$$(x', y') \oplus (x'', y'') = (x' + x'', y' + y'' + 2x'x''),$$

and

$$(x, y) \oplus ((x', y') \oplus (x'', y'')) = (x + (x' + x''), y + (y' + y'' + 2x'x'') + 2x(x' + x'')),$$

whose second coordinate is

$$y + y' + y'' + 2x'x'' + 2xx' + 2xx''.$$

The two second-coordinate expressions are identical (they both equal $y + y' + y'' + 2xx' + 2xx'' + 2x'x''$), and the first coordinates obviously agree. Thus associativity holds.

Identity. Take $(0, 0) \in \mathbb{R}^2$. For any (x, y)

$$(x, y) \oplus (0, 0) = (x + 0, y + 0 + 2x \cdot 0) = (x, y),$$

and similarly $(0, 0) \oplus (x, y) = (x, y)$. Hence $(0, 0)$ is the neutral element.

Inverse. Fix $(x, y) \in \mathbb{R}^2$ and seek (u, v) with

$$(x, y) \oplus (u, v) = (0, 0).$$

From the first coordinate $x + u = 0$ so $u = -x$. Substituting into the second coordinate equation

$$y + v + 2xu = 0 \implies v = -y - 2xu = -y - 2x(-x) = -y + 2x^2.$$

Thus the (two-sided) inverse of (x, y) is

$$(x, y)^{-1} = (-x, -y + 2x^2).$$

Verify it is also a left inverse:

$$(-x, -y + 2x^2) \oplus (x, y) = (-x + x, -y + 2x^2 + y + 2(-x)x) = (0, 0),$$

since $2(-x)x = -2x^2$ cancels $+2x^2$. Thus every element has an inverse.

Commutativity. For any $(x, y), (x', y')$,

$$(x, y) \oplus (x', y') = (x + x', y + y' + 2xx')$$

and

$$(x', y') \oplus (x, y) = (x' + x, y' + y + 2x'x') = (x + x', y + y' + 2x'x').$$

Since $2xx' = 2x'x'$, the two results are equal; hence $\oplus$ is commutative.

Having closure, associativity, an identity, inverses, and commutativity, $(\mathbb{R}^2, \oplus)$ is an abelian group.

2. Let $H = \{(x, x^2) \mid x \in \mathbb{R}\}$.

Nonempty / identity. $(0, 0) = (0, 0^2) \in H$, so $H \neq \emptyset$ and contains the neutral element.

Closure. Take arbitrary $(x, x^2), (x', x'^2) \in H$. Then

$$(x, x^2) \oplus (x', x'^2) = (x + x', x^2 + x'^2 + 2xx').$$

But

$$x^2 + x'^2 + 2xx' = (x + x')^2,$$

so

$$(x, x^2) \oplus (x', x'^2) = (x + x', (x + x')^2) \in H.$$

Inverses. The inverse in $(\mathbb{R}^2, \oplus)$ of a general element (x, y) is $(-x, -y + 2x^2)$. For $(x, x^2) \in H$ this becomes

$$(x, x^2)^{-1} = (-x, -x^2 + 2x^2) = (-x, x^2).$$

But $(-x, x^2) = (-x, (-x)^2) \in H$. Verify directly:

$$(x, x^2) \oplus (-x, x^2) = (x - x, x^2 + x^2 + 2x(-x)) = (0, 2x^2 - 2x^2) = (0, 0).$$

Thus inverses of elements of H lie in H .

Having shown H contains the identity, is closed under $\oplus$, and is closed under taking inverses, H is a subgroup of $(\mathbb{R}^2, \oplus)$.

3. Define $\Phi : \mathbb{R} \rightarrow H$ by $\Phi(t) = (t, t^2)$.

Homomorphism. For any $x, y \in \mathbb{R}$,

$$\Phi(x + y) = (x + y, (x + y)^2) = (x + y, x^2 + y^2 + 2xy).$$

Meanwhile

$$\Phi(x) \oplus \Phi(y) = (x, x^2) \oplus (y, y^2) = (x + y, x^2 + y^2 + 2xy).$$

Therefore $\Phi(x + y) = \Phi(x) \oplus \Phi(y)$ for all x, y , so Φ is a group homomorphism.

Injective. Suppose $\Phi(x) = \Phi(y)$. Then

$$(x, x^2) = (y, y^2).$$

Equality of ordered pairs implies equality of first coordinates, hence $x = y$. Thus Φ is injective.

Surjective onto H . By definition of H , every element $h \in H$ has the form $h = (t, t^2)$ for some $t \in \mathbb{R}$. For that t we have $\Phi(t) = h$. Hence Φ is surjective onto H .

Being a bijective homomorphism, Φ is an isomorphism from $(\mathbb{R}, +)$ onto $(H, \oplus)$.

Exercise 9 (Remarkable subgroups). Let $(G, \cdot)$ be a group, H and K be two subgroups of G , and $a \in G$. Show that the following sets are subgroups of G .

1. $H \cap K$.

2. $Z(G) = \{x \in G \mid \forall y \in G, xy = yx\}$ (the center of G).
3. $C_G(a) = \{x \in G \mid xa = ax\}$ (the centralizer of a in G).
4. $aHa^{-1} = \{aha^{-1} \mid h \in H\}$.
5. $N(H) = \{g \in G \mid gHg^{-1} = H\}$ (the normalizer of H in G).

Solution. We show for each set that it is nonempty and closed under the group operation and taking inverses.

1. *Intersection $H \cap K$.* Since H and K are subgroups, $e \in H$ and $e \in K$, so $e \in H \cap K$ and $H \cap K \neq \emptyset$. Take $x, y \in H \cap K$. Because H and K are subgroups, $xy \in H$ and $xy \in K$, hence $xy \in H \cap K$. Also $x^{-1} \in H$ and $x^{-1} \in K$, so $x^{-1} \in H \cap K$. Therefore $H \cap K$ is a subgroup.

2. *Center $Z(G)$.* The identity e commutes with every element, so $e \in Z(G)$. Let $x, y \in Z(G)$. For any $g \in G$ we have

$$(xy)g = x(yg) = x(gy) = (xg)y = (gx)y = g(xy),$$

so xy commutes with every g , hence $xy \in Z(G)$. If $x \in Z(G)$ then for every $g \in G$ we have $xg = gx$. Multiply on the left by x^{-1} and on the right by x^{-1} to obtain

$$g = x^{-1}gx \implies x^{-1}g = gx^{-1},$$

so x^{-1} also commutes with every g . Thus $x^{-1} \in Z(G)$. Hence $Z(G)$ is a subgroup.

3. *Centralizer $C_G(a)$.* Clearly $e \in C_G(a)$ since $ea = ae$. If $x, y \in C_G(a)$ then $xa = ax$ and $ya = ay$. Hence

$$(xy)a = x(ya) = x(ay) = (xa)y = (ax)y = a(xy),$$

so $xy \in C_G(a)$. If $x \in C_G(a)$ then $xa = ax$, and multiplying on the left by x^{-1} and on the right by x^{-1} gives $x^{-1}a = ax^{-1}$, so $x^{-1} \in C_G(a)$. Therefore $C_G(a)$ is a subgroup.

4. *Conjugate subgroup aHa^{-1} .* First note $aea^{-1} = e$, so $e \in aHa^{-1}$ and the set is nonempty. Take arbitrary elements aha^{-1} and $ah'a^{-1}$ in aHa^{-1} (with $h, h' \in H$). Their product is

$$(aha^{-1})(ah'a^{-1}) = a(hh')a^{-1},$$

and since $hh' \in H$ (because H is a subgroup), we have $a(hh')a^{-1} \in aHa^{-1}$. The inverse of aha^{-1} is

$$(aha^{-1})^{-1} = ah^{-1}a^{-1},$$

and $h^{-1} \in H$, so the inverse lies in aHa^{-1} . Thus aHa^{-1} is a subgroup of G .

5. *Normalizer $N(H)$.* Note $eHe^{-1} = H$, so $e \in N(H)$ and $N(H) \neq \emptyset$. Let $g_1, g_2 \in N(H)$. Then by definition $g_1Hg_1^{-1} = H$ and $g_2Hg_2^{-1} = H$. Consider

$$(g_1g_2)H(g_1g_2)^{-1} = g_1(g_2Hg_2^{-1})g_1^{-1} = g_1Hg_1^{-1} = H,$$

hence $g_1g_2 \in N(H)$. Finally, if $g \in N(H)$ then $gHg^{-1} = H$. Conjugating both sides by g^{-1} yields

$$g^{-1}Hg = H,$$

so $g^{-1} \in N(H)$. (Equivalently: from $gHg^{-1} = H$ multiply on the left by g^{-1} and on the right by g .) Therefore $N(H)$ is a subgroup of G .

Exercise 10.

1. Prove that the group $(\mathbb{R}, +)$ is not cyclic.

2. Show that any subgroup of a cyclic group is cyclic and give an example of a non-cyclic group whose proper subgroups are all cyclic.
3. Let p and q be two primes. Show that if G is a finite group of order pq , then every proper subgroup of G is cyclic.

Solution.

1. Suppose for contradiction that $(\mathbb{R}, +)$ is cyclic. Then there exists $a \in \mathbb{R}$ such that

$$\mathbb{R} = \langle a \rangle = \{na \mid n \in \mathbb{Z}\}.$$

The map

$$f : \mathbb{Z} \longrightarrow \langle a \rangle, \quad f(n) = na$$

is well-defined and bijective:

- Injective: If $f(n) = f(m)$ then $na = ma$, hence $(n - m)a = 0$. If $a \neq 0$ then $n - m = 0$, so $n = m$. If $a = 0$ then $\langle a \rangle = \{0\}$, which is not $\mathbb{R}$.
- Surjective: By definition of $\langle a \rangle$, every element has the form na for some $n \in \mathbb{Z}$, i.e. $f(n) = na$.

Therefore $\langle a \rangle$ is in bijection with $\mathbb{Z}$ and thus *countable*.

On the other hand, $\mathbb{R}$ is *uncountable*. One standard proof (Cantor's diagonal argument) is: Suppose $\mathbb{R}$ were countable. Then we could list all reals in $(0, 1)$ as $r_1, r_2, \dots$ with decimal expansions

$$r_1 = 0.d_{11}d_{12}d_{13}\dots, \quad r_2 = 0.d_{21}d_{22}d_{23}\dots, \quad \text{etc.}$$

Construct a new number $s = 0.c_1c_2c_3\dots$ by choosing each $c_i \neq d_{ii}$ and $c_i \in \{1, \dots, 8\}$ (avoid 0 or 9 to prevent ambiguity). Then s differs from r_i in the i -th decimal place, hence $s \notin \{r_1, r_2, \dots\}$ — a contradiction. Thus $(0, 1)$, and hence $\mathbb{R}$, is uncountable.

Since $\langle a \rangle$ is countable but $\mathbb{R}$ is uncountable, $\langle a \rangle \neq \mathbb{R}$. Therefore $(\mathbb{R}, +)$ is not cyclic.

2. Let $G = \langle g \rangle$ be cyclic and $H \leq G$. We show H is cyclic.

Case 1: G infinite. Then $G \cong (\mathbb{Z}, +)$ via $\varphi : \mathbb{Z} \rightarrow G$, $\varphi(k) = g^k$ (Check the textbook page 7). Every subgroup of $(\mathbb{Z}, +)$ has the form $d\mathbb{Z}$ for some $d \in \mathbb{N}$, generated by d . If $\varphi(d\mathbb{Z}) = H$ then $H = \langle g^d \rangle$ is cyclic.

Case 2: G finite of order n . Then $G \cong \mathbb{Z}/n\mathbb{Z}$ via $\varphi(\bar{k}) = g^k$ (here $\bar{k}$ denotes the residue class of k modulo n). Any subgroup of $\mathbb{Z}/n\mathbb{Z}$ has the form $\langle \bar{d} \rangle$ where $d \mid n$. If $\varphi(\langle \bar{d} \rangle) = H$ then $H = \langle g^d \rangle$ is cyclic.

In both cases, H is cyclic.

Example of a non-cyclic group whose proper subgroups are cyclic: The Klein four-group

$$V_4 = \{e, a, b, c\}, \quad a^2 = b^2 = c^2 = e, \quad ab = c, \quad bc = a, \quad ca = b,$$

has no element of order 4 (so it is not cyclic). Each nontrivial proper subgroup has order 2 and is cyclic, e.g. $\{e, a\}$, $\{e, b\}$, $\{e, c\}$.

3. Let G be finite with $|G| = pq$ where p, q are primes. By Lagrange's theorem, any proper subgroup $H < G$ has order dividing pq but strictly less than pq , so $|H| \in \{1, p, q\}$. If $|H| = 1$ it is trivial. If $|H|$ is prime, say $|H| = p$ or $|H| = q$, then any $h \in H \setminus \{e\}$ has order equal to $|H|$ (Lagrange), hence $\langle h \rangle = H$ and H is cyclic.

Therefore every proper subgroup of G is cyclic.

Exercise 11. Let H and K be two subgroups of a group G .

1. Show that $H \cup K$ is a subgroup of G if and only if $H \subset K$ or $K \subset H$.

2. We define HK by $HK = \{hk \mid h \in H, k \in K\}$. Show that HK is a subgroup of G if and only if $HK = KH$.

Solution.

- (1) *If direction.* If $H \subset K$ then $H \cup K = K$, which is a subgroup. Similarly if $K \subset H$ then $H \cup K = H$, a subgroup. Thus the condition is sufficient.

Only if direction. Suppose $H \cup K$ is a subgroup of G . We show one of H, K is contained in the other. Assume for contradiction that neither inclusion holds. Then there exist

$$h \in H \setminus K \quad \text{and} \quad k \in K \setminus H.$$

Since $H \cup K$ is a subgroup, it is closed under products; hence $hk \in H \cup K$.

- If $hk \in H$, then multiply on the left by $h^{-1} \in H$ to get

$$k = h^{-1}(hk) \in H,$$

contradicting $k \notin H$.

- If $hk \in K$, multiply on the right by $k^{-1} \in K$ to get

$$h = (hk)k^{-1} \in K,$$

contradicting $h \notin K$.

Both cases give contradictions, so our assumption was false: one of $H \subset K$ or $K \subset H$ must hold. This completes the proof of (1).

- (2) First note the easy inclusions $H \subset HK$ and $K \subset HK$ (take $e \in K$ or $e \in H$), so HK is nonempty.

($\Rightarrow$) *If HK is a subgroup, then $HK = KH$.* Assume HK is a subgroup. Since $K \subset HK$ and HK is a subgroup, every element of HK is closed under conjugation by elements of HK . In particular, for arbitrary $h \in H$ and $k \in K$ we have $hk \in HK$ and also $k \in HK$, hence

$$k(hk)k^{-1} = (kh) \in HK.$$

Thus every element $kh \in KH$ actually lies in HK , so $KH \subset HK$. By symmetry (interchanging the roles of H and K) we also get $HK \subset KH$. Therefore $HK = KH$.

($\Leftarrow$) *If $HK = KH$, then HK is a subgroup.* Assume $HK = KH$. We check the subgroup criteria.

- *Closure under product.* Let $x_1, h_1k_1 \in HK$ and $x_2 = h_2k_2 \in HK$. Then

$$x_1x_2 = (h_1k_1)(h_2k_2) = h_1(k_1h_2)k_2.$$

Since $k_1h_2 \in KH$ and $KH = HK$, we can write $k_1h_2 = h_3k_3$ for some $h_3 \in H, k_3 \in K$. Thus

$$x_1x_2 = h_1(h_3k_3)k_2 = (h_1h_3)(k_3k_2) \in HK,$$

so $x_1x_2 \in HK$.

- *Closure under inverses.* Let $x = hk \in HK$. Then

$$x^{-1} = (hk)^{-1} = k^{-1}h^{-1}.$$

As $k^{-1}h^{-1} \in KH$ and $KH = HK$, we have $x^{-1} \in HK$.

Since HK is nonempty and closed under products and inverses, it is a subgroup of G .

Combining both directions yields: HK is a subgroup of G if and only if $HK = KH$.

Exercise 12 (Classification of subgroups of $(\mathbb{R}, +)$). Let G be a subgroup of $(\mathbb{R}, +)$ such that $G \neq \{0\}$. We will show that either G is dense in $\mathbb{R}$ or there exists $\alpha \in \mathbb{R}_{>0}$ with $G = \alpha\mathbb{Z}$.

1. Show that the set $S := G \cap \mathbb{R}_{>0}$ is nonempty.
2. Put $\alpha = \inf S$ and suppose $\alpha > 0$.
 - (a) Prove that $\alpha \in G$ and deduce $\alpha\mathbb{Z} \subset G$.
 - (b) Show that every $g \in G$ can be written as $g = \alpha \lfloor \frac{g}{\alpha} \rfloor$, hence $G \subset \alpha\mathbb{Z}$.
 - (c) Conclude that $G = \alpha\mathbb{Z}$.
3. Suppose $\alpha = \inf S = 0$.
 - (a) Show that for all real $x < y$ there exists $g \in G$ with $x < g < y$.
 - (b) Deduce that G is dense in $\mathbb{R}$.

Complete solution.

(1) Since $G \neq \{0\}$ there exists $t \in G$ with $t \neq 0$. If $t > 0$ then $t \in S$ and we are done. If $t < 0$ then $-t \in G$ and $-t > 0$, so $-t \in S$. Hence $S \neq \emptyset$.

(2) Let $\alpha = \inf S$ and assume $\alpha > 0$.

(a) *Every positive element of G is an integer multiple of α .* Fix $s \in S$. Define the integer $q := \lfloor \frac{s}{\alpha} \rfloor \geq 1$, so

$$q\alpha \leq s < (q+1)\alpha.$$

Set $r := s - q\alpha$. Then $r \in G$ (since $s \in G$ and $q\alpha \in \mathbb{R}$; more precisely later we will show integer multiples of α are in G — for the moment note $r \geq 0$ and $r < \alpha$). But $0 \leq r < \alpha$; if $r > 0$ then $r \in S$ contradicts that α is the greatest lower bound of S . Thus $r = 0$, so

$$s = q\alpha.$$

Therefore every $s \in S$ is of the form $q\alpha$ with $q \in \mathbb{N}$.

Existence of α in G . Pick any $s \in S$, then $s = q\alpha$ for some $q \in \mathbb{N}$. Consider the set of positive integers

$$Q := \{q \in \mathbb{N} \mid q\alpha \in S\}.$$

By the previous paragraph $Q \neq \emptyset$. Let $m = \min Q$. Then $m\alpha \in S$. If $m > 1$ then $(m-1)\alpha = m\alpha - \alpha$ would be a positive number smaller than $m\alpha$ and, since $m\alpha \in G$ and α is the infimum of positive elements of G , one checks that this forces $(m-1)\alpha \in S$, contradicting the minimality of m . Hence $m = 1$ and so $\alpha \in S \subset G$. (Thus the infimum is actually achieved.)

Consequently $\alpha \in G$. Since G is a subgroup, all integer multiples of α belong to G , i.e. $\alpha\mathbb{Z} \subset G$.

(b) Now let $g \in G$ be arbitrary. Using the (real) division algorithm with divisor $\alpha > 0$ we write

$$g = q\alpha + r, \quad q := \left\lfloor \frac{g}{\alpha} \right\rfloor \in \mathbb{Z}, \quad 0 \leq r < \alpha.$$

Because g and $q\alpha$ lie in G , their difference $r = g - q\alpha$ also lies in G . If $r > 0$ then $r \in S$ and $r < \alpha$, contradicting that α is the infimum of S . Thus $r = 0$ and $g = q\alpha$. Therefore every $g \in G$ is an integer multiple of α , so $G \subset \alpha\mathbb{Z}$.

(c) Combining (a) and (b) we get $G = \alpha\mathbb{Z}$, as required.

(3) Finally suppose $\alpha = \inf S = 0$. We prove G is dense in $\mathbb{R}$.

(a) Let $x < y$ be real numbers. Since $\alpha = 0$ is the infimum of S , for any $\varepsilon > 0$ there exists $g \in S$ with $0 < g < \varepsilon$. Take $\varepsilon := y - x$. Choose $0 < g < y - x$ with $g \in G$. Now choose an integer n such that

$$x < ng \leq x + g < x + (y - x) = y.$$

(We can take $n = \lfloor \frac{x}{g} \rfloor + 1$.) Because $g \in G$ and $n \in \mathbb{Z}$ we have $ng \in G$, and by construction $x < ng < y$. Thus there exists $g' \in G$ in the interval (x, y) .

(b) Since every open interval contains an element of G , the subgroup G is dense in $\mathbb{R}$.

This completes the classification: any nonzero subgroup $G \leq (\mathbb{R}, +)$ is either of the form $\alpha\mathbb{Z}$ for some $\alpha > 0$, or is dense in $\mathbb{R}$.

Exercise 13 (On the subgroups of S^1). Let n be a positive integer. Denote

$$U = \{z \in \mathbb{C} \mid |z| = 1\} \quad (\text{the unit circle}), \quad U_n = \{z \in \mathbb{C} \mid z^n = 1\} \quad (\text{the } n\text{-th roots of unity}),$$

and

$$U_\infty = \bigcup_{n \geq 1} U_n \quad (\text{the set of all roots of unity}).$$

I.

1. Show that U and U_n are subgroups of $(\mathbb{C}^\times, \times)$. Is this the case for U_∞ ?
2. Show that $U_n \subset U_m$ if and only if $n \mid m$.
3. Show that U_n is cyclic.

II. Let $\alpha \in \mathbb{R}$ and define for fixed $n \geq 1$ the map

$$f_n : \mathbb{Z} \longrightarrow \mathbb{C}^\times, \quad f_n(p) = e^{2\pi i p n \alpha}.$$

1. Show that f_n is a group homomorphism from $(\mathbb{Z}, +)$ to $(\mathbb{C}^\times, \times)$.
2. Show that $\text{Im } f_n \subset U$.
3. Show that f_n is injective if and only if $\alpha \notin \mathbb{Q}$.

III. Consider the map

$$\varphi : \mathbb{R} \longrightarrow U, \quad \varphi(t) = e^{it}.$$

1. Show that φ is a group homomorphism from $(\mathbb{R}, +)$ to $(U, \times)$. Is φ injective? Surjective?
2. Define a relation $\mathcal{R}$ on $\mathbb{R}$ by

$$\forall x, y \in \mathbb{R}, \quad x \mathcal{R} y \iff \exists k \in \mathbb{Z} \text{ such that } x - y = 2k\pi.$$

Show that $\mathcal{R}$ is an equivalence relation.

3. On the quotient set $\mathbb{R}/\mathcal{R}$ define

$$[x] \oplus [y] := [x + y].$$

Show that $(\mathbb{R}/\mathcal{R}, \oplus)$ is a group isomorphic to $(U, \times)$.

Solution.

I.1. U and U_n are nonempty (they contain 1). Take $z, w \in U$. Then $|zw| = |z||w| = 1 \cdot 1 = 1$, so $zw \in U$. Also $|z^{-1}| = 1$, so $z^{-1} \in U$. Thus U is a subgroup of $(\mathbb{C}^\times, \times)$.

For U_n : if $z^n = 1$ and $w^n = 1$ then $(zw)^n = z^n w^n = 1 \cdot 1 = 1$, so $zw \in U_n$. If $z^n = 1$ then $(z^{-1})^n = (z^n)^{-1} = 1^{-1} = 1$, so $z^{-1} \in U_n$. Hence U_n is a subgroup.

For $U_\infty = \bigcup_{n \geq 1} U_n$: if $\zeta \in U_n$ and $\eta \in U_m$ then $(\zeta\eta)^{\text{lcm}(n,m)} = \zeta^{\text{lcm}(n,m)} \eta^{\text{lcm}(n,m)} = 1 \cdot 1 = 1$, so $\zeta\eta$ is a root of unity (indeed belongs to $U_{\text{lcm}(n,m)}$). Also inverses of roots of unity are roots of unity. Therefore U_∞ is closed under multiplication and inversion, so U_∞ is a subgroup of $\mathbb{C}^\times$. (In fact U_∞ is the torsion subgroup of U .)

I.2. If $n \mid m$ write $m = kn$. If $z \in U_n$, then $z^m = (z^n)^k = 1^k = 1$, so $z \in U_m$; thus $U_n \subset U_m$.

Conversely, assume $U_n \subset U_m$. Let $\zeta = e^{2\pi i/n}$ be a primitive n -th root of unity (so $\zeta \in U_n$ and $\zeta^k \neq 1$ for $1 \leq k < n$). By assumption $\zeta \in U_m$, so $\zeta^m = 1$. The order of ζ is n , so n divides m . Hence $n \mid m$.

I.3. The element $\zeta_n := e^{2\pi i/n}$ satisfies $\zeta_n^n = 1$ and its powers $\{1, \zeta_n, \zeta_n^2, \dots, \zeta_n^{n-1}\}$ are distinct, so they give exactly n elements of U_n . Therefore $U_n = \langle \zeta_n \rangle$ is cyclic of order n .

II.1. For $p, q \in \mathbb{Z}$,

$$f_n(p+q) = e^{2\pi i(p+q)n\alpha} = e^{2\pi ipn\alpha} e^{2\pi iq n\alpha} = f_n(p)f_n(q),$$

so f_n is a group homomorphism $(\mathbb{Z}, +) \rightarrow (\mathbb{C}^\times, \times)$.

II.2. For any $p \in \mathbb{Z}$,

$$|f_n(p)| = |e^{2\pi ipn\alpha}| = 1,$$

hence $f_n(p) \in U$. Thus $\text{Im } f_n \subset U$.

II.3. The kernel of f_n is

$$\ker f_n = \{p \in \mathbb{Z} \mid e^{2\pi ipn\alpha} = 1\} = \{p \in \mathbb{Z} \mid pn\alpha \in \mathbb{Z}\}.$$

If $\alpha \in \mathbb{Q}$ say $\alpha = a/b$ in lowest terms (with $a \in \mathbb{Z}, b \in \mathbb{N}$), then

$$pn\alpha = pn\frac{a}{b} \in \mathbb{Z} \iff b \mid pna.$$

Since b and a are coprime, this condition is equivalent to $b \mid pn$, so there exist nonzero integers p in the kernel (for example $p = b$ if $\gcd(b, n) = 1$ or an appropriate multiple otherwise). Thus $\ker f_n \neq \{0\}$ and f_n is not injective.

If $\alpha \notin \mathbb{Q}$ (i.e. α irrational) and $pn\alpha \in \mathbb{Z}$ for some integer p , then $\alpha = \frac{k}{pn} \in \mathbb{Q}$, contradiction unless $p = 0$. Hence $\ker f_n = \{0\}$ and f_n is injective. Therefore f_n is injective $\iff \alpha \notin \mathbb{Q}$.

III.1. For $s, t \in \mathbb{R}$,

$$\varphi(s+t) = e^{i(s+t)} = e^{is}e^{it} = \varphi(s)\varphi(t),$$

so φ is a group homomorphism $(\mathbb{R}, +) \rightarrow (U, \times)$.

The kernel is

$$\ker \varphi = \{t \in \mathbb{R} \mid e^{it} = 1\} = \{2k\pi \mid k \in \mathbb{Z}\} = 2\pi\mathbb{Z},$$

so φ is not injective. φ is surjective because every $z \in U$ can be written as $z = e^{i\theta}$ for some $\theta \in [0, 2\pi)$; thus there exists $t = \theta$ with $\varphi(t) = z$. Hence φ is surjective.

III.2. The relation $\mathcal{R}$ defined by $x\mathcal{R}y \iff x - y = 2k\pi$ for some $k \in \mathbb{Z}$ is an equivalence relation:

- Reflexive: $x - x = 0 = 2 \cdot 0\pi$. - Symmetric: if $x - y = 2k\pi$ then $y - x = -2k\pi = 2(-k)\pi$. - Transitive: if $x - y = 2k\pi$ and $y - z = 2\ell\pi$ then $x - z = 2(k + \ell)\pi$.

Thus it is an equivalence relation; denote classes by $[x]$.

III.3. Define $\oplus$ on $\mathbb{R}/\mathcal{R}$ by $[x] \oplus [y] := [x + y]$. This is well-defined because if $x' \sim x$ and $y' \sim y$ then $x' - x = 2k\pi$, $y' - y = 2\ell\pi$ so $(x' + y') - (x + y) = 2(k + \ell)\pi$ and hence $[x' + y'] = [x + y]$. With this operation $(\mathbb{R}/\mathcal{R}, \oplus)$ is a group: identity $[0]$, inverse of $[x]$ is $[-x]$, associativity follows from that of $+$ on $\mathbb{R}$.

Finally the surjective homomorphism $\varphi : \mathbb{R} \rightarrow U$ factors through the quotient: there is a unique bijective homomorphism $\tilde{\varphi} : \mathbb{R}/\mathcal{R} \rightarrow U$ such that $\varphi = \tilde{\varphi} \circ \pi$, where $\pi : \mathbb{R} \rightarrow \mathbb{R}/\mathcal{R}$ is the quotient map. Explicitly $\tilde{\varphi}([t]) = e^{it}$. Because $\ker \varphi = 2\pi\mathbb{Z}$, the induced map $\tilde{\varphi}$ is injective, and since φ is surjective, $\tilde{\varphi}$ is bijective. Hence $\tilde{\varphi}$ is an isomorphism between $(\mathbb{R}/\mathcal{R}, \oplus)$ and $(U, \times)$.

Exercise 14. On $\mathbb{R}$, we define the binary operation $*$ by

$$\forall x, y \in \mathbb{R}, \quad x * y = \sqrt[3]{x^3 + y^3}.$$

Show that $(\mathbb{R}, *)$ is an abelian group isomorphic to $(\mathbb{R}, +)$.

Solution.

Define the map

$$\Phi : (\mathbb{R}, *) \longrightarrow (\mathbb{R}, +), \quad \Phi(x) = x^3.$$

We verify that Φ is a group isomorphism.

Homomorphism. For any $x, y \in \mathbb{R}$,

$$\Phi(x * y) = (x * y)^3 = (\sqrt[3]{x^3 + y^3})^3 = x^3 + y^3 = \Phi(x) + \Phi(y),$$

hence $\Phi(x * y) = \Phi(x) + \Phi(y)$ and Φ is a homomorphism.

Bijectivity. The map $t \mapsto t^3$ is strictly increasing and continuous on $\mathbb{R}$, with range $\mathbb{R}$, therefore it is bijective. Thus Φ is bijective and has an inverse $\Phi^{-1}(u) = \sqrt[3]{u}$.

Since Φ is a bijective homomorphism, it is an isomorphism. Consequently $(\mathbb{R}, *)$ is a group and isomorphic to $(\mathbb{R}, +)$.

Explicit group properties from the isomorphism.

- *Identity:* Let e be the neutral element for $*$. Then $\Phi(e) = 0$ (neutral of $(\mathbb{R}, +)$), so $e^3 = 0$ and $e = 0$. Indeed $x * 0 = \sqrt[3]{x^3 + 0} = x$ for all x .
- *Inverse:* For $x \in \mathbb{R}$ the inverse x^{-1} satisfies $x * x^{-1} = 0$, hence $x^3 + (x^{-1})^3 = 0$, so $(x^{-1})^3 = -x^3$ and $x^{-1} = -x$. Thus the inverse of x is $-x$.
- *Associativity and commutativity:* These follow from the isomorphism with $(\mathbb{R}, +)$, or can be checked directly using the identity

$$(x * y) * z = \sqrt[3]{(x^3 + y^3) + z^3} = \sqrt[3]{x^3 + (y^3 + z^3)} = x * (y * z),$$

and

$$x * y = \sqrt[3]{x^3 + y^3} = \sqrt[3]{y^3 + x^3} = y * x.$$

Therefore $(\mathbb{R}, *)$ is an abelian group isomorphic to $(\mathbb{R}, +)$, as required.

Exercise 15.

- (I) (a) Determine all group homomorphisms $\varphi : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$. Which of these are injective? surjective?
- (b) Determine all group homomorphisms $\psi : (\mathbb{Q}, +) \rightarrow (\mathbb{Z}, +)$.
- (II) Prove the following statements.
- (a) There is no group isomorphism between $(\mathbb{R}, +)$ and $(\mathbb{R}^\times, \times)$.
- (b) There is no group isomorphism between $(\mathbb{Q}, +)$ and $(\mathbb{Q}_+^\times, \times)$.

Solution.

I.1. Homomorphisms $\mathbb{Z} \rightarrow \mathbb{Z}$.

Let $\varphi : (\mathbb{Z}, +) \rightarrow (\mathbb{Z}, +)$ be a group homomorphism. The group $(\mathbb{Z}, +)$ is generated by 1, so φ is determined by $k := \varphi(1) \in \mathbb{Z}$. For any $n \in \mathbb{Z}$,

$$\varphi(n) = \varphi(1 + \cdots + 1) = n\varphi(1) = kn.$$

Thus every homomorphism φ is "multiplication by k ":

$$\varphi_k : \mathbb{Z} \rightarrow \mathbb{Z}, \quad \varphi_k(n) = kn, \quad k \in \mathbb{Z},$$

and conversely each φ_k is a homomorphism.

Kernel and injectivity. $\ker \varphi_k = \{n \in \mathbb{Z} \mid kn = 0\}$. If $k = 0$ then $\ker \varphi_0 = \mathbb{Z}$ (map is zero, not injective). If $k \neq 0$, then $kn = 0$ implies $n = 0$, so $\ker \varphi_k = \{0\}$. Hence φ_k is injective $\iff k \neq 0$.

Image and surjectivity. $\text{Im } \varphi_k = k\mathbb{Z}$. Therefore φ_k is surjective $\iff k\mathbb{Z} = \mathbb{Z} \iff k = \pm 1$.

I.2. Homomorphisms $\mathbb{Q} \rightarrow \mathbb{Z}$.

Let $\psi : (\mathbb{Q}, +) \rightarrow (\mathbb{Z}, +)$ be a homomorphism. Set $a := \psi(1) \in \mathbb{Z}$. For any integer n we have $\psi(n) = na$ as before. Now for a rational p/q (with $p \in \mathbb{Z}, q \in \mathbb{N}_{>0}$) the homomorphism property gives

$$q\psi\left(\frac{p}{q}\right) = \psi\left(q \cdot \frac{p}{q}\right) = \psi(p) = pa.$$

Thus $\psi\left(\frac{p}{q}\right) = \frac{pa}{q}$. But $\psi\left(\frac{p}{q}\right)$ must be an integer for every choice of p, q . Taking $p = 1$ we get $a/q \in \mathbb{Z}$ for every $q \geq 1$, which forces $a = 0$. Hence ψ is the zero map. Conversely the zero map is a homomorphism. Therefore the only group homomorphism $\mathbb{Q} \rightarrow \mathbb{Z}$ is the trivial map; in particular there are no nontrivial injective or surjective homomorphisms.

II.1. No isomorphism between $(\mathbb{R}, +)$ and $(\mathbb{R}^\times, \times)$.

If two groups are isomorphic, they have the same group-theoretic invariants. Observe:

- The additive group $(\mathbb{R}, +)$ is torsion-free: the only element of finite order is 0. Indeed, if $x \in \mathbb{R}$ and $nx = 0$ for some positive integer n , then $x = 0$.
- The multiplicative group $(\mathbb{R}^\times, \times)$ contains -1 with $(-1)^2 = 1$, so -1 is an element of order 2 (a nontrivial torsion element).

Therefore $(\mathbb{R}, +)$ has no nontrivial element of finite order while $(\mathbb{R}^\times, \times)$ does; hence they cannot be isomorphic.

II.2. No isomorphism between $(\mathbb{Q}, +)$ and $(\mathbb{Q}_+^\times, \times)$.

Here $\mathbb{Q}_+^\times$ denotes the multiplicative group of positive rationals.

Two useful structural properties distinguish these groups.

(i) $(\mathbb{Q}, +)$ is *divisible*. For every $q \in \mathbb{Q}$ and every integer $n \geq 1$ there exists $r \in \mathbb{Q}$ with $nr = q$ (take $r = q/n$). A divisible abelian group has the property that for every element and every integer $n \geq 1$ the equation $nx = y$ is solvable in the group.

(ii) $\mathbb{Q}_+^\times$ is *not divisible*. For example, the equation $x^2 = 2$ has no solution in $\mathbb{Q}_+^\times$ (because $\sqrt{2} \notin \mathbb{Q}$), so 2 is an element of $\mathbb{Q}_+^\times$ that is not a square in $\mathbb{Q}_+^\times$. Thus there exists $y \in \mathbb{Q}_+^\times$ (namely $y = 2$) and $n = 2$ for which the equation $x^n = y$ has no solution in $\mathbb{Q}_+^\times$.

Divisibility is preserved by group isomorphism: an isomorphic image of a divisible group is divisible. Since $(\mathbb{Q}, +)$ is divisible but $(\mathbb{Q}_+^\times, \times)$ is not, they cannot be isomorphic.

(An alternative invariant: $(\mathbb{Q}, +)$ is a vector space over $\mathbb{Q}$ of dimension 1 (hence divisible), while $\mathbb{Q}_+^\times$ is isomorphic to a direct sum $\bigoplus_{p \text{ prime}} \mathbb{Z}$ (exponent vectors of primes) and therefore not divisible.)

This completes the solution.

Exercise 16. Let $(G, *)$ be a group and $\text{Aut}(G)$ be the set of automorphisms of G .

1. Show that $(\text{Aut}(G), \circ)$ is a group (composition of maps).
2. Let H be a subgroup of $\text{Aut}(G)$, and define

$$\phi : G \longrightarrow \mathcal{P}(G), \quad \phi(x) = \{f(x) \mid f \in H\}.$$

The set $\phi(x)$ is called the *orbit* of x under H . Show that the collection $\phi(G) = \{\phi(x) \mid x \in G\}$ is a partition of G .

Solution.

1. We check the group axioms for $(\text{Aut}(G), \circ)$, where $\circ$ denotes composition of maps.

Closure. If $f, g \in \text{Aut}(G)$ then f and g are bijective homomorphisms $G \rightarrow G$. The composition $f \circ g$ is a homomorphism (composition of homomorphisms) and bijective (composition of bijections), hence $f \circ g \in \text{Aut}(G)$.

Associativity. Composition of maps is associative: for any $f, g, h \in \text{Aut}(G)$,

$$(f \circ g) \circ h = f \circ (g \circ h).$$

Identity. The identity map $\text{id}_G : G \rightarrow G$, $\text{id}_G(x) = x$, is a bijective homomorphism, so $\text{id}_G \in \text{Aut}(G)$. For any $f \in \text{Aut}(G)$,

$$f \circ \text{id}_G = f = \text{id}_G \circ f.$$

Inverses. If $f \in \text{Aut}(G)$ then f is a bijective homomorphism, so its inverse map $f^{-1} : G \rightarrow G$ exists and is a homomorphism (the inverse of a bijective homomorphism is a homomorphism). Thus $f^{-1} \in \text{Aut}(G)$ and $f \circ f^{-1} = \text{id}_G = f^{-1} \circ f$.

Hence all group axioms hold, and $(\text{Aut}(G), \circ)$ is a group.

2. Let $H \leq \text{Aut}(G)$ and define the relation $\sim$ on G by

$$x \sim y \iff \exists f \in H \text{ such that } y = f(x).$$

We show that $\sim$ is an equivalence relation; its equivalence classes are precisely the sets $\phi(x)$, and the family of equivalence classes is a partition of G .

Reflexive. Since H is a subgroup of $\text{Aut}(G)$ it contains the identity map id_G . For every $x \in G$, $x = \text{id}_G(x)$, so $x \sim x$.

Symmetric. If $x \sim y$ then $y = f(x)$ for some $f \in H$. Because $f \in H \leq \text{Aut}(G)$, we have $f^{-1} \in H$. Applying f^{-1} gives $x = f^{-1}(y)$, hence $y \sim x$.

Transitive. If $x \sim y$ and $y \sim z$ then $y = f(x)$ and $z = g(y)$ for some $f, g \in H$. The composition $g \circ f$ lies in H (closure of subgroup under composition), and $z = (g \circ f)(x)$, so $x \sim z$.

Thus $\sim$ is an equivalence relation. For $x \in G$ the equivalence class of x is

$$[x]_{\sim} = \{ y \in G \mid y = f(x) \text{ for some } f \in H \} = \phi(x).$$

Equivalence classes are nonempty, pairwise disjoint, and their union equals G . Therefore the family $\phi(G) = \{\phi(x) \mid x \in G\}$ is exactly the set of equivalence classes of $\sim$, hence a partition of G .

RINGS

Exercise 17. Complete the operation tables for the ring $R = \{a, b, c, d\}$.

+	a	b	c	d	*	a	b	c	d
a	a	b	c	d	a	a	a	a	a
b	b	a	d	c	b	a	b		
c	c	d	a	b	c	a		a	
d	d	c	b	a	d	a	b	c	

Hint. Use distributivity together with the relation $d = b + c$ (which follows from the addition table).

Solution. From the addition table one has the identities $b + b = a$, $c + c = a$, $d + d = a$ and $b + c = d$, $b + d = c$, $c + d = b$. Using $d = b + c$ and distributivity,

$$d * b = (b + c) * b = b * b + c * b.$$

The partial multiplication table gives $d * b = b$ and $b * b = b$, hence $b = b + c * b$. Adding b to both sides (recalling $b + b = a$) yields $a = c * b$. Thus $c * b = a$. Using the same distributive decompositions and the given entries one deduces $b * c = a$, $c * c = c$, and $b * d = b$. Finally, expanding $d * d = (b + c) * (b + c)$ and substituting the computed products yields

$$d * d = b * b + b * c + c * b + c * c = b + a + a + c = b + c = d.$$

Collecting the computed entries produces the completed multiplication table below (the final, consistent ring table).

$*$	a	b	c	d
a	a	a	a	a
b	a	b	a	b
c	a	a	c	c
d	a	b	c	d

Characteristic. The characteristic of a ring R is the smallest positive integer n such that $n \cdot 1 = 0$ (if it exists), where $n \cdot 1$ denotes the n -fold sum $1 + \cdots + 1$. Here the multiplicative identity is d , and $2 \cdot d = d + d = a$ is the additive neutral element; therefore $\text{char}(R) = 2$.

Remarks. The ring is commutative, unital with unity d , its additive group is the Klein four-group, and every nonzero element is idempotent ($x^2 = x$ for $x \in \{b, c, d\}$).

Exercise 18. Let $(R, +, \times)$ be a commutative ring, and let N be the set of nilpotent elements of R . Put

$$B = \{1 + x \mid x \in N\}.$$

Show that $(B, \times)$ is a group.

Solution. Since 0 is nilpotent, $1 = 1 + 0 \in B$, so B is nonempty and contains the multiplicative identity of R .

If $x, y \in N$ then $(1 + x)(1 + y) = 1 + (x + y + xy)$. We claim $z := x + y + xy$ is nilpotent. Choose integers $n, m \geq 1$ with $x^n = 0$ and $y^m = 0$. Expanding z^{n+m} and grouping factors using commutativity, every monomial is of the form $x^r y^s$ with $r + s \geq n + m$; hence either $r \geq n$ or $s \geq m$, so each monomial is zero. Therefore $z^{n+m} = 0$ and $z \in N$. Thus $(1 + x)(1 + y) = 1 + z \in B$, so B is closed under multiplication.

Finally, for $x \in N$ choose $N \geq 1$ with $x^N = 0$. The finite geometric series

$$(1 + x)^{-1} = 1 - x + x^2 - \cdots + (-1)^{N-1} x^{N-1}$$

is an element of R and satisfies $(1 + x)(1 + x)^{-1} = 1 - (-1)^N x^N = 1$, so it is the multiplicative inverse of $1 + x$. Since this inverse is of the form $1 + y$ with y a polynomial in x without constant term, we have $y \in N$ and hence $(1 + x)^{-1} \in B$.

Therefore B contains the identity, is closed under multiplication, and contains inverses, so $(B, \times)$ is a group (indeed a subgroup of the unit group $R^\times$).

Exercise 19. Let $\mathbb{Z}/n\mathbb{Z}$ be the set of integers modulo n .

I.

1. Verify that $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring.
2. What is the characteristic of the ring $\mathbb{Z}/5\mathbb{Z}$? and the ring $\mathbb{Z}/6\mathbb{Z}$?
3. Show that $\mathbb{Z}/n\mathbb{Z}$ is an integral domain if and only if n is prime.

II. Let m and n be two integers. Put $m\mathbb{Z} + n\mathbb{Z} = \{an + bm \mid a, b \in \mathbb{Z}\}$. Show that:

1. $m\mathbb{Z} + n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$.
2. m and $n + 7m$ generate the subgroup $m\mathbb{Z} + n\mathbb{Z}$.

III. Let m and n be two positive integers.

1. Show that $m\mathbb{Z} \cap n\mathbb{Z} = \text{lcm}(m, n)\mathbb{Z}$.
2. Is $2\mathbb{Z} \cup 3\mathbb{Z}$ a subring of $\mathbb{Z}$?
3. Find a necessary and sufficient condition on m and n for which $m\mathbb{Z} \cup n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$.
4. Determine the smallest subgroup of $\mathbb{Z}$ containing $m\mathbb{Z}$ and $n\mathbb{Z}$.

Solution.

I.1. Represent elements of $\mathbb{Z}/n\mathbb{Z}$ by residue classes $\bar{k}$ ($k \in \mathbb{Z}$). Define

$$\bar{a} + \bar{b} = \overline{a+b}, \quad \bar{a} \cdot \bar{b} = \overline{ab}.$$

These operations are well-defined: if $\bar{a} = \bar{a'}$ and $\bar{b} = \bar{b'}$ then $a - a' = qn$, $b - b' = rn$ for integers q, r , so

$$(a+b) - (a'+b') = (q+r)n, \quad ab - a'b' = (a-a')b' + (b-b')a = (qn)b' + (rn)a$$

are multiples of n , hence $\overline{a+b} = \overline{a'+b'}$ and $\overline{ab} = \overline{a'b'}$. The ring axioms (associativity, commutativity of $+$ and $\cdot$, distributivity, existence of additive identity $\bar{0}$ and additive inverses $-\bar{a} = \overline{-a}$) are inherited from $\mathbb{Z}$. Thus $\mathbb{Z}/n\mathbb{Z}$ is a commutative ring; its multiplicative identity is $\bar{1}$.

I.2. The characteristic of $\mathbb{Z}/n\mathbb{Z}$ is the smallest positive t with $t \cdot \bar{1} = \bar{0}$. But $t \cdot \bar{1} = \bar{t}$, so the least such t is n . Hence

$$\text{char}(\mathbb{Z}/5\mathbb{Z}) = 5, \quad \text{char}(\mathbb{Z}/6\mathbb{Z}) = 6.$$

I.3. ($\Rightarrow$) Suppose that n is not prime and $\mathbb{Z}/n\mathbb{Z}$ is an integral domain, write $n = ab$ with $1 < a, b < n$. Then $\bar{a} \neq \bar{0}$ and $\bar{b} \neq \bar{0}$ but $\bar{a} \cdot \bar{b} = \overline{ab} = \bar{n} = \bar{0}$, so there are zero divisors which contradicts the hypothesis. This proves the desired implication.

($\Leftarrow$) Suppose n is prime and let $\bar{a}, \bar{b} \in \mathbb{Z}/n\mathbb{Z}$ with $\bar{a}\bar{b} = \bar{0}$ in $\mathbb{Z}/n\mathbb{Z}$. This means ab is divisible by n . Since n is prime, for ab divisible by n to hold, at least one of a or b must be divisible by n . Hence, there are no nonzero divisors. Thus, $\mathbb{Z}/n\mathbb{Z}$ is an integral domain.

Therefore $\mathbb{Z}/n\mathbb{Z}$ is an integral domain iff n is prime.

II.1. Let $H = m\mathbb{Z} + n\mathbb{Z}$. Clearly $0 \in H$ (take $a = b = 0$). If $x = an + bm$ and $y = a'n + b'm$ are in H then

$$x - y = (a - a')n + (b - b')m \in H,$$

so H is closed under subtraction and hence under addition and additive inverses. Thus H is a subgroup of $(\mathbb{Z}, +)$.

II.2. Let $H' = \langle m, n + 7m \rangle$ be the subgroup generated by m and $n + 7m$. Since

$$n + 7m = 1 \cdot n + 7 \cdot m \in m\mathbb{Z} + n\mathbb{Z},$$

we have $H' \subseteq m\mathbb{Z} + n\mathbb{Z}$. Conversely any generator n of $m\mathbb{Z} + n\mathbb{Z}$ can be expressed from m and $n + 7m$:

$$n = (n + 7m) - 7m,$$

so $n \in H'$. Hence $m\mathbb{Z} + n\mathbb{Z} \subseteq H'$. Therefore $H' = m\mathbb{Z} + n\mathbb{Z}$, i.e. m and $n + 7m$ generate the same subgroup.

III.1. If $d = \text{lcm}(m, n)$ then d is a multiple of m and of n , so $d\mathbb{Z} \subseteq m\mathbb{Z} \cap n\mathbb{Z}$. Conversely, let $x \in m\mathbb{Z} \cap n\mathbb{Z}$. Then $x = um = vn$ for some integers u, v , so x is a common multiple of m and n , hence $d \mid x$. Thus $x \in d\mathbb{Z}$. Therefore $m\mathbb{Z} \cap n\mathbb{Z} = d\mathbb{Z}$.

III.2. $2\mathbb{Z} \cup 3\mathbb{Z}$ is not a subring of $\mathbb{Z}$ because it is not closed under addition: $2 \in 2\mathbb{Z}$ and $3 \in 3\mathbb{Z}$, but $2 + 3 = 5 \notin 2\mathbb{Z} \cup 3\mathbb{Z}$. Hence it is not a subring.

III.3. By the subgroup-union criterion (same as for subgroups of an abelian group), $m\mathbb{Z} \cup n\mathbb{Z}$ is a subgroup of $\mathbb{Z}$ iff one of the two is contained in the other, i.e. iff $m\mathbb{Z} \subseteq n\mathbb{Z}$ or $n\mathbb{Z} \subseteq m\mathbb{Z}$. Now $m\mathbb{Z} \subseteq n\mathbb{Z}$ holds exactly when $n \mid m$. Thus the necessary and sufficient condition is that $m \mid n$ or $n \mid m$.

III.4. The smallest subgroup of $\mathbb{Z}$ containing both $m\mathbb{Z}$ and $n\mathbb{Z}$ is their sum

$$m\mathbb{Z} + n\mathbb{Z} = \{an + bm \mid a, b \in \mathbb{Z}\}.$$

It is a standard fact that $m\mathbb{Z} + n\mathbb{Z} = \text{gcd}(m, n)\mathbb{Z}$ (one proves this using Bézout's identity: there exist $u, v \in \mathbb{Z}$ with $um + vn = \text{gcd}(m, n)$, so $\text{gcd}(m, n) \in m\mathbb{Z} + n\mathbb{Z}$, and any element of $m\mathbb{Z} + n\mathbb{Z}$ is a multiple of $\text{gcd}(m, n)$). Hence the smallest subgroup containing $m\mathbb{Z}$ and $n\mathbb{Z}$ is $\text{gcd}(m, n)\mathbb{Z}$.

Exercise 20. We define on $\mathbb{C}$ the binary operations $\oplus$ and $\otimes$ as follows. For $z = a + ib$, $z' = a' + ib'$ (with $a, b, a', b' \in \mathbb{R}$) set

$$z \oplus z' = (a + a') + i(b + b'), \quad z \otimes z' = aa' + i(ab' + a'b).$$

1. Show that $(\mathbb{C}, \oplus, \otimes)$ is a commutative ring.
2. Determine the set U of invertible elements (units) in $(\mathbb{C}, \oplus, \otimes)$.
3. Determine the set D of (nonzero) zero divisors in $(\mathbb{C}, \oplus, \otimes)$.
4. Let $z = a + ib \in \mathbb{C}$. For $n \in \mathbb{N}$ define $z^{(n)}$ (the n -fold $\otimes$ -product)

$$z^{(n)} = \underbrace{z \otimes z \otimes \cdots \otimes z}_{n \text{ times}}.$$

- (a) Express $z^{(n)}$ in terms of n, a, b .
- (b) Solve in $\mathbb{C}$ the equation $z^{(4)} \oplus z^{(2)} = 6 + i$.

Solution.

It is convenient to identify $z = a + ib$ with the ordered pair $(a, b) \in \mathbb{R}^2$. Under this identification

$$(a, b) \oplus (a', b') = (a + a', b + b'), \quad (a, b) \otimes (a', b') = (aa', ab' + a'b).$$

All verifications are done in this pair notation.

1. $(\mathbb{C}, \oplus)$ is clearly an abelian group: $\oplus$ is componentwise addition, neutral element $(0, 0)$, and inverse of (a, b) is $(-a, -b)$. The operation $\otimes$ is closed on $\mathbb{R}^2$. For $(a, b), (a', b'), (a'', b'') \in \mathbb{R}^2$ compute

$$((a, b) \otimes (a', b')) \otimes (a'', b'') = (aa', ab' + a'b) \otimes (a'', b'') = (aa'a'', aa'b'' + (ab' + a'b)a''),$$

while

$$(a, b) \otimes ((a', b') \otimes (a'', b'')) = (a, b) \otimes (a'a'', a'b'' + a''b') = (aa'a'', a(a'b'' + a''b') + b(a'a'')).$$

Using commutativity and associativity of real multiplication these two second-coordinates are equal: $aa'b'' + ab'a'' + a'ba'' = aa'b'' + aa'b'' + ba'a''$. Hence $\otimes$ is associative. Commutativity of $\otimes$ follows from $aa' = a'a$ and $ab' + a'b = a'b + ab'$. Distributivity holds because

$$\begin{aligned} (a, b) \otimes ((a', b') \oplus (a'', b'')) &= (a, b) \otimes (a' + a'', b' + b'') = (a(a' + a''), a(b' + b'') + b(a' + a'')) \\ &= (aa' + aa'', ab' + ab'' + ba' + ba'') = (aa', ab' + a'b) \oplus (aa'', ab'' + a''b), \end{aligned}$$

which equals $(a, b) \otimes (a', b') \oplus (a, b) \otimes (a'', b'')$. Thus $(\mathbb{C}, \oplus, \otimes)$ is a commutative ring with multiplicative identity $(1, 0)$ (i.e. $1 + 0i$).

2. Units. Let $z = (a, b)$. If z has an inverse $w = (x, y)$ for $\otimes$ then

$$(a, b) \otimes (x, y) = (1, 0),$$

which gives the system

$$ax = 1, \quad ay + bx = 0.$$

The first equation forces $a \neq 0$ and $x = 1/a$. Then $ay + b(1/a) = 0$ gives $y = -\frac{b}{a^2}$. Thus an inverse exists iff $a \neq 0$. Therefore the group of units is

$$U = \{a + ib \in \mathbb{C} \mid a \neq 0\}.$$

Explicitly, for $a \neq 0$,

$$(a, b)^{-1} = \left(\frac{1}{a}, -\frac{b}{a^2}\right), \quad \text{equivalently} \quad (a + ib)^{-1} = \frac{1}{a} - i \frac{b}{a^2}.$$

If $a = 0$ then $ax = 0$ for all x and cannot equal 1; hence no inverse exists when $a = 0$.

3. Zero divisors. A (nonzero) element $z = (a, b)$ is a zero divisor iff there exists a nonzero $w = (c, d)$ with $(a, b) \otimes (c, d) = (0, 0)$. This yields

$$ac = 0, \quad ad + bc = 0.$$

If $a \neq 0$ then $ac = 0$ implies $c = 0$, and then $ad + bc = ad = 0$ gives $d = 0$, so $w = 0$. Thus no nonzero zero-divisor can have $a \neq 0$. Conversely if $a = 0$ and $b \neq 0$ then for any $w = (0, d)$ one has $(0, b) \otimes (0, d) = (0, 0)$ while both factors can be nonzero. In particular each nonzero element of the form $(0, b)$ (pure imaginary in our notation) is a (nonzero) zero divisor and in fact is nilpotent:

$$(0, b) \otimes (0, b) = (0, 0).$$

Therefore the set of nonzero zero divisors is

$$D = \{a + ib \in \mathbb{C} \mid a = 0, b \neq 0\}.$$

(Zero itself also satisfies $0 \otimes w = 0$ for all w , but conventionally when listing zero divisors one usually emphasizes nonzero.)

4.(a) Powers $z^{(n)}$. Let $z = a + ib$.

For $n = 1$: $z^{(1)} = a + ib$.

For $n = 2$:

$$z^{(2)} = (a + ib) \otimes (a + ib) = a^2 + i(ab + ab) = a^2 + i(2ab).$$

For $n = 3$:

$$z^{(3)} = z^{(2)} \otimes z = (a^2 + i(2ab)) \otimes (a + ib) = a^3 + i(a^2b + 2ab \cdot a) = a^3 + i(3a^2b).$$

For $n = 4$:

$$z^{(4)} = z^{(3)} \otimes z = (a^3 + i(3a^2b)) \otimes (a + ib) = a^4 + i(a^3b + 3a^3b) = a^4 + i(4a^3b).$$

From these computations we guess the general pattern:

$$z^{(n)} = a^n + i(n a^{n-1}b) \quad \text{for all } n \geq 1.$$

Proof by induction. For $n = 1$ the formula is trivial. Suppose it holds for n . Then

$$(a, b)^{(n+1)} = (a, b)^{(n)} \otimes (a, b) = (a^n, n a^{n-1}b) \otimes (a, b) = (a^n a, a^n b + (n a^{n-1}b)a).$$

The real coordinate is a^{n+1} ; the imaginary coordinate is $a^n b + n a^n b = (n+1)a^n b$. This proves the formula. (If $a = 0$ then for $n \geq 2$ the right-hand side becomes $(0, 0)$, matching direct computation: $(0, b)^{(2)} = (0, 0)$, etc.)

Translating back to complex notation,

$$z^{(n)} = a^n + i(n a^{n-1}b).$$

4.(b) Solve $z^{(4)} \oplus z^{(2)} = 6 + i$. Using the formula

$$z^{(4)} = a^4 + i 4a^3b, \quad z^{(2)} = a^2 + i 2ab,$$

the $\oplus$ -sum yields

$$(a^4 + a^2) + i(4a^3b + 2ab) = 6 + i.$$

Equating real and imaginary parts gives the system

$$a^4 + a^2 = 6, \quad ab(4a^2 + 2) = 1.$$

Set $x = a^2 \geq 0$. The first equation becomes $x^2 + x - 6 = 0$, i.e. $(x+3)(x-2) = 0$. Since $x \geq 0$ we get $x = 2$, hence $a^2 = 2$ and $a = \pm\sqrt{2}$.

Substitute $a^2 = 2$ into the second equation:

$$ab(4 \cdot 2 + 2) = ab \cdot 10 = 1 \implies b = \frac{1}{10a}.$$

Therefore the solutions are

$$a = \sqrt{2}, \quad b = \frac{1}{10\sqrt{2}}, \quad \text{or} \quad a = -\sqrt{2}, \quad b = -\frac{1}{10\sqrt{2}}.$$

In complex form the two solutions are

$$z = \sqrt{2} + i\frac{1}{10\sqrt{2}} \quad \text{and} \quad z = -\sqrt{2} - i\frac{1}{10\sqrt{2}}.$$

These are all solutions (no other real a satisfy the real equation, and $a = 0$ is impossible since it would make the real part 0).

Exercise 21. On $\mathbb{R}^2$ consider componentwise addition

$$(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$$

and the multiplication

$$(x_1, x_2) \times (y_1, y_2) = (x_1y_1 - x_2y_2, x_1y_2 + x_2y_1).$$

1. Verify that $(\mathbb{R}^2, +, \times)$ is a ring.
2. Find the zero divisors of $(\mathbb{R}^2, +, \times)$.

Solution.

Identify $\mathbb{R}^2$ with the complex numbers via the bijection

$$\Phi : \mathbb{R}^2 \longrightarrow \mathbb{C}, \quad \Phi(x_1, x_2) = x_1 + ix_2.$$

Under this identification the given addition and multiplication on $\mathbb{R}^2$ are exactly the usual complex addition and multiplication:

$$\Phi((x_1, x_2) + (y_1, y_2)) = \Phi(x_1, x_2) + \Phi(y_1, y_2),$$

$$\Phi((x_1, x_2) \times (y_1, y_2)) = \Phi(x_1, x_2) \cdot \Phi(y_1, y_2).$$

Since $(\mathbb{C}, +, \cdot)$ is a commutative ring with unit, all ring axioms (associativity and commutativity of $+$, existence of additive identity $(0, 0)$ and additive inverses $-(x_1, x_2) = (-x_1, -x_2)$, associativity and commutativity of $\times$, distributivity of $\times$ over $+$, and existence of multiplicative identity $(1, 0)$) hold for $(\mathbb{R}^2, +, \times)$. Thus $(\mathbb{R}^2, +, \times)$ is a commutative unital ring.

It is instructive (and elementary) to verify associativity and distributivity directly from the coordinate formulas, but the identification with $\mathbb{C}$ makes this immediate.

To determine zero divisors, let $(x_1, x_2), (y_1, y_2) \in \mathbb{R}^2$ satisfy

$$(x_1, x_2) \times (y_1, y_2) = (0, 0),$$

i.e.

$$x_1y_1 - x_2y_2 = 0, \quad x_1y_2 + x_2y_1 = 0.$$

Square and add these two equations to obtain

$$(x_1y_1 - x_2y_2)^2 + (x_1y_2 + x_2y_1)^2 = (x_1^2 + x_2^2)(y_1^2 + y_2^2) = 0.$$

Since $x_1^2 + x_2^2 \geq 0$ and $y_1^2 + y_2^2 \geq 0$, the product is zero iff at least one factor is zero. Thus

$$x_1^2 + x_2^2 = 0 \quad \text{or} \quad y_1^2 + y_2^2 = 0.$$

But $x_1^2 + x_2^2 = 0$ (resp. $y_1^2 + y_2^2 = 0$) holds exactly when $(x_1, x_2) = (0, 0)$ (resp. $(y_1, y_2) = (0, 0)$). Therefore whenever a product is zero one of the factors must be the zero vector. Hence there are no nonzero zero divisors.

Consequently the only zero divisor is $(0, 0)$ itself; equivalently, $(\mathbb{R}^2, +, \times)$ is an integral domain (indeed it is isomorphic to the field $\mathbb{C}$).

Exercise 22. Let $R = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$. For $x = a + b\sqrt{2} \in R$ define the norm

$$N(x) = a^2 - 2b^2.$$

1. Show that R is a subring of $\mathbb{R}$.
2. Show that for all $x, y \in R$ one has $N(xy) = N(x)N(y)$.
3. Deduce that x is invertible in R if and only if $N(x) = \pm 1$.
4. Show that the elements $\pm(1 + \sqrt{2})^n$ ($n \in \mathbb{Z}$) are units of R .
5. Conversely show that every unit of R is of the form $\pm(1 + \sqrt{2})^n$.
 - (a) Show that we may assume a unit x has the form $x = a + b\sqrt{2}$ with $a \in \mathbb{N}$, $b \in \mathbb{N}_0$.
 - (b) Show that such an x equals $(1 + \sqrt{2})^n$ for some $n \in \mathbb{N}_0$.

Solution.

1. R is a unital subring of $\mathbb{R}$. We verify the subring criterion you requested: (i) R is an additive subgroup of $(\mathbb{R}, +)$; (ii) R is closed under multiplication; (iii) $1 \in R$.

Let $x = a + b\sqrt{2}$ and $y = c + d\sqrt{2}$ with $a, b, c, d \in \mathbb{Z}$. Then

$$x + y = (a + c) + (b + d)\sqrt{2} \in R, \quad -x = (-a) + (-b)\sqrt{2} \in R,$$

so R is an additive subgroup. Also

$$xy = (ac + 2bd) + (ad + bc)\sqrt{2} \in R,$$

so R is closed under multiplication. Finally $1 = 1 + 0\sqrt{2} \in R$. Therefore R is a unital subring of $\mathbb{R}$.

2. *Multiplicativity of the norm.* Write $x = a + b\sqrt{2}$, $y = c + d\sqrt{2}$. Then

$$xy = (ac + 2bd) + (ad + bc)\sqrt{2},$$

and a direct expansion yields

$$N(xy) = (ac + 2bd)^2 - 2(ad + bc)^2 = (a^2 - 2b^2)(c^2 - 2d^2) = N(x)N(y).$$

Equivalently, with conjugation $\overline{a + b\sqrt{2}} = a - b\sqrt{2}$ one has $N(x) = x\bar{x}$ and $N(xy) = (xy)\overline{(xy)} = x\bar{x}y\bar{y} = N(x)N(y)$.

3. *Units characterized by the norm.* If $x \in R$ is a unit then there exists $y \in R$ with $xy = 1$. Applying N gives

$$1 = N(1) = N(x)N(y),$$

and since $N(x), N(y) \in \mathbb{Z}$ we obtain $N(x) = \pm 1$. Conversely, if $N(x) = \pm 1$ then $\bar{x} \in R$ and

$$x \cdot \frac{\bar{x}}{N(x)} = \frac{x\bar{x}}{N(x)} = \frac{N(x)}{N(x)} = 1,$$

so $x^{-1} = \frac{\bar{x}}{N(x)} \in R$. Thus x is a unit in R iff $N(x) = \pm 1$.

4. The elements $\pm(1 + \sqrt{2})^n$ are units. Compute $N(1 + \sqrt{2}) = 1^2 - 2 \cdot 1^2 = -1$. By multiplicativity,

$$N((1 + \sqrt{2})^n) = (N(1 + \sqrt{2}))^n = (-1)^n \in \{\pm 1\},$$

so each $(1 + \sqrt{2})^n$ is a unit. Multiplying by -1 preserves invertibility, hence every $\pm(1 + \sqrt{2})^n$ is a unit.

5. Every unit is of the form $\pm(1 + \sqrt{2})^n$. Let $x = a + b\sqrt{2} \in R^\times$ with integers a, b and $N(x) = \pm 1$. We first reduce to the case $a \geq 0, b \geq 0$: replacing x by $-x$ or x^{-1} (both operations preserve the statement to be proved up to sign and inversion) we may assume $a \geq 0$ and $b \geq 0$.

If $b = 0$ then $x = \pm 1$ and the claim holds. Suppose $b \geq 1$. Perform one division by $1 + \sqrt{2}$ in $\mathbb{Q}(\sqrt{2})$:

$$\frac{x}{1 + \sqrt{2}} = \frac{a + b\sqrt{2}}{1 + \sqrt{2}} = \frac{(a + b\sqrt{2})(1 - \sqrt{2})}{(1 + \sqrt{2})(1 - \sqrt{2})} = -(a + b\sqrt{2})(1 - \sqrt{2}).$$

Set

$$x_1 := -\frac{x}{1 + \sqrt{2}} = (a + b\sqrt{2})(1 - \sqrt{2}) = (2b - a) + (a - b)\sqrt{2}.$$

Put $a_1 := 2b - a$ and $b_1 := a - b$. From $N(x) = a^2 - 2b^2 = \pm 1$ one finds a^2 is very close to $2b^2$; in particular for $b \geq 1$ one checks $b < a < 2b$, hence $a_1 = 2b - a \geq 0$ and $b_1 = a - b \geq 0$ with $b_1 < b$. Moreover $x_1 \in R$ and

$$N(x_1) = \frac{N(x)}{N(1 + \sqrt{2})} = \pm 1,$$

so x_1 is again a unit.

Thus this division step reduces the second coefficient strictly: from b to $b_1 < b$. Iterating produces a finite sequence of units $x, x_1, x_2, \dots$ whose second coefficients strictly decrease until one obtains some x_n with second coefficient 0. Such x_n equals ± 1 . Unwinding the divisions shows that x is obtained from ± 1 by multiplying (and possibly changing sign) by $1 + \sqrt{2}$ a finite number of times; hence $x = \pm(1 + \sqrt{2})^n$ for some $n \in \mathbb{N}_0$. Allowing inverses yields the full description with $n \in \mathbb{Z}$.

Therefore

$$R^\times = \{\pm(1 + \sqrt{2})^n \mid n \in \mathbb{Z}\},$$

as required.

Exercise 23. Let

$$R = \left\{ \frac{a}{10^n} \mid a \in \mathbb{Z}, n \in \mathbb{N} \right\}.$$

1. Show that R is a subring of $\mathbb{Q}$ for the usual binary operations.
2. Show that 3 has no inverse in R . Conclude.
3. Determine the set of invertible elements of R .

Solution.

1. R is a unital subring of $\mathbb{Q}$. Let $x = \frac{a}{10^n}$ and $y = \frac{b}{10^m}$ be arbitrary elements of R (with $a, b \in \mathbb{Z}, n, m \in \mathbb{N}$). Put $t = \max\{n, m\}$. Writing both fractions with common denominator 10^t we have

$$x = \frac{a10^{t-n}}{10^t}, \quad y = \frac{b10^{t-m}}{10^t},$$

so the sum

$$x + y = \frac{a10^{t-n} + b10^{t-m}}{10^t}$$

has integer numerator and denominator 10^t , hence $x + y \in R$. The additive inverse $-x = -\frac{a}{10^n}$ clearly lies in R . For the product,

$$xy = \frac{ab}{10^{n+m}} \in R,$$

since $ab \in \mathbb{Z}$ and $n + m \in \mathbb{N}$. Finally $1 = 1/10^0 \in R$. Thus R is an additive subgroup of $\mathbb{Q}$, closed under multiplication and containing 1; hence R is a unital subring of $\mathbb{Q}$.

2. *3 has no inverse in R .* Suppose, for contradiction, that 3 has an inverse in R . Then there exist integers a and $n \geq 0$ with

$$\frac{a}{10^n} \cdot 3 = 1 \implies 3a = 10^n.$$

But the right-hand side $10^n = 2^n 5^n$ is divisible only by the primes 2 and 5, while the left-hand side is divisible by 3; no integer equality of this form is possible. Therefore 3 has no inverse in R . In particular R is not a field.

3. *Description of the group of units $R^\times$.* Let $u = \frac{a}{10^n} \in R$ be nonzero. Then u is invertible in R iff there is $v = \frac{b}{10^m} \in R$ with

$$uv = \frac{ab}{10^{n+m}} = 1 \implies ab = 10^{n+m} = 2^{n+m} 5^{n+m}.$$

Thus ab is a power of 10. This forces every prime divisor of a to lie among $\{2, 5\}$; equivalently $a = \pm 2^r 5^s$ for some integers $r, s \geq 0$. Conversely, if $a = \pm 2^r 5^s$ then choose $k \geq r + s$ and put $b = \pm 2^{k-r} 5^{k-s}$ and $m = k - (n)$ (adjusting m so that $n + m = k$); then $ab = 10^k$ and $v = b/10^m \in R$ is an inverse of u . Hence u is invertible in R precisely when the integer numerator a has no prime factors other than 2 and 5.

Equivalently, every unit of R can be written (uniquely) as

$$\pm 2^\alpha 5^\beta \quad \text{with } \alpha, \beta \in \mathbb{Z}.$$

Indeed any invertible element $\frac{a}{10^n}$ with $a = \pm 2^r 5^s$ equals $\pm 2^{r-n} 5^{s-n}$, and conversely any $\pm 2^\alpha 5^\beta$ (with $\alpha, \beta \in \mathbb{Z}$) lies in R and is invertible (its inverse is $\pm 2^{-\alpha} 5^{-\beta}$, also of the same form).

Therefore the group of units of R is

$$R^\times = \{ \pm 2^\alpha 5^\beta \mid \alpha, \beta \in \mathbb{Z} \},$$

i.e. the set of nonzero rationals whose numerator and denominator (in lowest terms) involve only the primes 2 and 5.

(As a remark: this is the localization of $\mathbb{Z}$ at the multiplicative set generated by 2 and 5; equivalently $R = \mathbb{Z}[1/10]$ and $R^\times = \{\pm 2^\alpha 5^\beta\}$.)

Exercise 24.

1. Let R be a ring and $C = \{x \in R \mid \forall y \in R, xy = yx\}$ (the *center* of R). Show that C is a subring of R .
2. Determine the smallest subring of $\mathbb{Q}$ that contains $\frac{1}{2}$.
3. (a) Find all ring homomorphisms $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/30\mathbb{Z}$.
(b) For each homomorphism, describe its kernel and its image.

Solution.

1. *C is a subring of R .* We check the subring criterion (additive subgroup, closed under multiplication, contains 1 when R is unital).

First $0 \in C$ and $1 \in C$ (if R has a unity) because 0 and 1 commute with every element. Let $x, x' \in C$. For any $y \in R$,

$$(x + x')y = xy + x'y = yx + yx' = y(x + x'),$$

so $x + x' \in C$. Also

$$(-x)y = -(xy) = -(yx) = y(-x),$$

so $-x \in C$. Thus C is an additive subgroup of R . Finally, for $x, x' \in C$ and any $y \in R$,

$$(xx')y = x(x'y) = x(yx') = (xy)x' = (yx)x' = y(xx'),$$

so $xx' \in C$. Therefore C is closed under multiplication. If R has 1 then $1 \in C$. Hence C is a (unital) subring of R .

2. *Smallest subring of $\mathbb{Q}$ containing $\frac{1}{2}$.* Any subring $S \subseteq \mathbb{Q}$ containing $\frac{1}{2}$ must contain $\mathbb{Z}$ (since it is an additive subgroup) and must be closed under multiplication, so it contains all powers $(\frac{1}{2})^k = \frac{1}{2^k}$ for $k \geq 0$. Hence it contains every finite $\mathbb{Z}$ -linear combination of these elements, i.e. all rationals of the form

$$\frac{m}{2^k} \quad (m \in \mathbb{Z}, k \in \mathbb{N}_0).$$

The set

$$\mathbb{Z}[\frac{1}{2}] = \left\{ \frac{m}{2^k} \mid m \in \mathbb{Z}, k \in \mathbb{N}_0 \right\}$$

is easily checked to be a subring of $\mathbb{Q}$ containing $\frac{1}{2}$, and by the previous paragraph it is contained in every subring of $\mathbb{Q}$ that contains $\frac{1}{2}$. Therefore $\mathbb{Z}[\frac{1}{2}]$ is the smallest subring of $\mathbb{Q}$ containing $\frac{1}{2}$.

3.(a) *Ring homomorphisms $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/30\mathbb{Z}$.* A ring homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/30\mathbb{Z}$ is determined by $e := \varphi(1)$. For any $n \in \mathbb{Z}$,

$$\varphi(n) = n\varphi(1) = ne \pmod{30}.$$

Compatibility with multiplication requires $e^2 = \varphi(1)^2 = \varphi(1) = e$ in $\mathbb{Z}/30\mathbb{Z}$; hence e must be an idempotent modulo 30.

Idempotents modulo 30 are exactly the solutions of $e^2 \equiv e \pmod{30}$. Using the Chinese Remainder Theorem,

$$\mathbb{Z}/30\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z},$$

and idempotents in the product ring are exactly the tuples with each component equal to 0 or 1. Thus there are $2^3 = 8$ idempotents. Concretely the idempotent residues modulo 30 are

$$e \in \{0, 1, 6, 10, 15, 16, 21, 25\},$$

(they are the integers e with $0 \leq e < 30$ satisfying $e^2 \equiv e \pmod{30}$). For each choice of such an e the formula $\varphi(n) = ne \pmod{30}$ defines a ring homomorphism $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}/30\mathbb{Z}$. Thus there are exactly eight ring homomorphisms.

3.(b) *Kernels and images.* Fix an idempotent e and write $g = \gcd(e, 30)$. The image of φ is the ideal of $\mathbb{Z}/30\mathbb{Z}$ generated by e ; in $\mathbb{Z}/30\mathbb{Z}$ every principal ideal is generated by a divisor of 30, and one checks

$$\text{Im } \varphi = e \cdot (\mathbb{Z}/30\mathbb{Z}) = g \cdot (\mathbb{Z}/30\mathbb{Z}) = \{0, g, 2g, \dots, (30/g - 1)g\} \subseteq \mathbb{Z}/30\mathbb{Z}.$$

The kernel consists of integers $n \in \mathbb{Z}$ such that $ne \equiv 0 \pmod{30}$. Writing $e = ge'$ and $30 = gt$ with $\gcd(e', t) = 1$, the condition $30 \mid ne$ is equivalent to $t \mid n$. Hence

$$\ker \varphi = t\mathbb{Z} = \frac{30}{g} \mathbb{Z}.$$

For the eight idempotents above the pairs $(e, g = \gcd(e, 30), \ker, \text{Im})$ are:

e	$g = \gcd(e, 30)$	$\ker \varphi$	$\text{Im } \varphi$
0	30	$1 \cdot 30\mathbb{Z} = 30\mathbb{Z}$	$\{0\}$
1	1	$30\mathbb{Z}/1\mathbb{Z}$ (i.e. $30\mathbb{Z}$)	$\mathbb{Z}/30\mathbb{Z}$
6	6	$\frac{30}{6}\mathbb{Z} = 5\mathbb{Z}$	$6\mathbb{Z}/30\mathbb{Z}$
10	10	$3\mathbb{Z}$	$10\mathbb{Z}/30\mathbb{Z}$
15	15	$2\mathbb{Z}$	$15\mathbb{Z}/30\mathbb{Z}$
16	2	$15\mathbb{Z}$	$2\mathbb{Z}/30\mathbb{Z}$
21	3	$10\mathbb{Z}$	$3\mathbb{Z}/30\mathbb{Z}$
25	5	$6\mathbb{Z}$	$5\mathbb{Z}/30\mathbb{Z}$

(Here by $d\mathbb{Z}/30\mathbb{Z}$ we mean the ideal of $\mathbb{Z}/30\mathbb{Z}$ generated by d , i.e. the set $\{0, d, 2d, \dots\}$.)

In words: for the homomorphism with $\varphi(1) = e$, the kernel is $\frac{30}{\gcd(e, 30)}\mathbb{Z}$ and the image is the ideal of $\mathbb{Z}/30\mathbb{Z}$ generated by $\gcd(e, 30)$.

Exercise 25 (Particular ideals). Let R be a ring with unity.

1. The sum of two ideals I and J is the set defined by $I + J = \{i + j \mid i \in I, j \in J\}$. Show that $I + J$ is also an ideal.
2. Suppose that R is commutative, and let I and J be two ideals of R . Define

$$[I : J] = \{r \in R \mid \forall s \in J, rs \in I\}.$$

Show that $[I : J]$ is an ideal (the *quotient ideal* of I by J).

3. Suppose that R is commutative.

- (a) Let J be an ideal of R . The *radical* of J is

$$\sqrt{J} = \{a \in R \mid a^n \in J \text{ for some } n \in \mathbb{N}\}.$$

Show that $\sqrt{J}$ is an ideal of R .

- (b) Deduce that the set of nilpotent elements

$$I = \{x \in R \mid x^n = 0 \text{ for some } n \in \mathbb{N}\}$$

(the nilradical) is an ideal of R .

- (c) Let $A \subseteq R$. The *annihilator* of A is

$$\text{Ann}(A) = \{r \in R \mid ra = 0 \text{ for all } a \in A\}.$$

Show that $\text{Ann}(A)$ is an ideal of R .

Solution.

1. Let I, J be ideals of R . We show $I + J$ is an ideal.

First $0 = 0 + 0 \in I + J$. If $x = i_1 + j_1$ and $y = i_2 + j_2$ with $i_1, i_2 \in I$ and $j_1, j_2 \in J$, then

$$x - y = (i_1 - i_2) + (j_1 - j_2) \in I + J,$$

since I and J are additive subgroups. Thus $I + J$ is an additive subgroup.

Next let $r \in R$ and $x = i + j \in I + J$. Then

$$rx = ri + rj \in I + J$$

because I and J are ideals (hence closed under multiplication by arbitrary ring elements). Similarly $xr \in I + J$. Therefore $I + J$ is a two-sided ideal of R .

2. Let R be commutative and I, J ideals. Put

$$[I : J] = \{r \in R \mid \forall s \in J, rs \in I\}.$$

We prove $[I : J]$ is an ideal.

First $0 \in [I : J]$ because $0 \cdot s = 0 \in I$ for all $s \in J$. If $r_1, r_2 \in [I : J]$ then for every $s \in J$,

$$(r_1 - r_2)s = r_1s - r_2s \in I - I = I,$$

so $r_1 - r_2 \in [I : J]$. Thus $[I : J]$ is an additive subgroup.

Let $r \in [I : J]$ and $t \in R$. For every $s \in J$ we have

$$(tr)s = t(rs).$$

Since $rs \in I$ and I is an ideal, $t(rs) \in I$. Hence $(tr)s \in I$ for all $s \in J$, so $tr \in [I : J]$. Commutativity was used only to simplify left/right distinctions; the same argument shows $[I : J]$ is a two-sided ideal when R is commutative (indeed the definition already uses multiplication on the left). Therefore $[I : J]$ is an ideal of R .

3.(a) Let J be an ideal of the commutative ring R and set

$$\sqrt{J} = \{a \in R \mid a^n \in J \text{ for some } n \geq 1\}.$$

We show $\sqrt{J}$ is an ideal.

If $a \in \sqrt{J}$ then $a^n \in J$ for some n . For any $r \in R$ we have $(ra)^n = r^n a^n \in J$ because $a^n \in J$ and J is an ideal; hence $ra \in \sqrt{J}$. Also $(-a)^n = (-1)^n a^n \in J$, so $-a \in \sqrt{J}$. It remains to check closure under addition.

Let $a, b \in \sqrt{J}$ with $a^n \in J$ and $b^m \in J$. Consider $N = n + m$. By the binomial theorem (using commutativity),

$$(a + b)^N = \sum_{k=0}^N \binom{N}{k} a^k b^{N-k}.$$

Every term in this sum has either exponent of a at least n (if $k \geq n$) or exponent of b at least m (if $N - k \geq m$). Hence each term lies in J (because it contains a factor a^n or b^m), and so the whole sum $(a + b)^N \in J$. Thus $a + b \in \sqrt{J}$. Therefore $\sqrt{J}$ is an ideal.

3.(b) The set of nilpotent elements

$$I = \{x \in R \mid x^n = 0 \text{ for some } n \geq 1\}$$

is exactly $\sqrt{(0)}$, the radical of the zero ideal. By part (a) $\sqrt{(0)}$ is an ideal, hence the nilradical I is an ideal.

3.(c) Let $A \subseteq R$ and $\text{Ann}(A) = \{r \in R \mid ra = 0 \forall a \in A\}$. We show $\text{Ann}(A)$ is an ideal (no commutativity required).

Clearly $0 \in \text{Ann}(A)$. If $r_1, r_2 \in \text{Ann}(A)$ and $a \in A$ then

$$(r_1 - r_2)a = r_1a - r_2a = 0 - 0 = 0,$$

so $r_1 - r_2 \in \text{Ann}(A)$; hence $\text{Ann}(A)$ is an additive subgroup.

If $r \in \text{Ann}(A)$ and $t \in R$, then for every $a \in A$,

$$(tr)a = t(ra) = t \cdot 0 = 0,$$

and similarly

$$(rt)a = r(ta).$$

Thus tr and rt annihilate every element of A , so both lie in $\text{Ann}(A)$. Therefore $\text{Ann}(A)$ is a two-sided ideal of R .

Exercise 26 (Boolean ring). Let R be a ring in which every element is idempotent: $\forall x \in R, x^2 = x$.

1. Give examples of such rings.
2. Show that for every $a \in R$ we have $2a = 0$.
3. Deduce that R is commutative.
4. Show that R cannot have exactly three elements.
5. Assume R is finite and $|R| > 2$. Show that R has zero divisors. (Hint: consider $xy(x + y)$.)

6. Show that if $|R| = 4$ then R is unique up to isomorphism.
7. Show that if R is finite then $|R|$ is a power of 2.

Solution.

1. *Examples.* The simplest examples are the field $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ and finite direct products of copies of $\mathbb{F}_2$, for instance $\mathbb{F}_2 \times \mathbb{F}_2$. Another natural example is the power set algebra $\mathcal{P}(X)$ of any set X with symmetric difference as addition and intersection as multiplication: for $A \subseteq X$ we have $A \cap A = A$, so every element is idempotent. All these are Boolean rings.

2. $2a = 0$. For arbitrary $a \in R$ compute

$$0 = (a + a)^2 - (a + a) = (a^2 + a^2 + aa + aa) - (a + a).$$

Using $a^2 = a$ this simplifies to

$$0 = (a + a + aa + aa) - (a + a) = aa + aa = 2(aa).$$

But $aa = a$, so $2a = 0$. A quicker standard computation is

$$0 = (a + a)^2 - (a + a) = a^2 + a^2 + aa + aa - (a + a) = aa + aa - (a + a) = 2a - 2a = 0,$$

from which $2a = 0$ follows; in any case the conclusion is the same: $2a = 0$ for all $a \in R$. Hence the characteristic of R is 2 (or divides 2).

3. *Commutativity.* Expand $(x + y)^2 = x + y$. Using idempotence and distributivity,

$$x + y = (x + y)^2 = x^2 + xy + yx + y^2 = x + xy + yx + y.$$

Cancelling $x + y$ (possible since addition is group subtraction) yields

$$xy + yx = 0.$$

But from part (2) we have $2xy = 0$, hence

$$xy = -yx = yx,$$

so multiplication is commutative. Thus every Boolean ring is commutative.

4. *No ring of cardinal 3.* From (2) the additive group $(R, +)$ has exponent 2: every element satisfies $2a = 0$. Hence $(R, +)$ is a vector space over $\mathbb{F}_2$, so its cardinality is a power of 2. In particular it cannot be 3. Equivalently, a nonzero subring must contain the infinite subgroup $\{n \cdot 1 : n \in \mathbb{Z}\}$ unless $1 + 1 = 0$; here $1 + 1 = 0$ so the additive group order is a power of 2, not 3.

5. *Existence of zero divisors when $|R| > 2$.* If $|R| > 2$ there exists an element x with $x \notin \{0, 1\}$. Consider $x(1 - x)$. Since $x \neq 0$ and $1 - x \neq 0$ (otherwise $x = 1$ or $x = 0$), compute

$$x(1 - x) = x - x^2 = x - x = 0.$$

Thus both x and $1 - x$ are nonzero and their product is zero: they are zero divisors. (The hint polynomial $xy(x + y)$ gives the same idea when taking $y = 1 - x$ or more generally.)

6. *Uniqueness for $|R| = 4$.* Let R be a Boolean ring with $|R| = 4$. By (2) the additive group is a vector space over $\mathbb{F}_2$; hence $(R, +) \cong \mathbb{F}_2^2$. Choose an $\mathbb{F}_2$ -basis $\{e_1, e_2\}$ of $(R, +)$ with $e_1 = 1$ or, if 1 is not in the chosen basis, include 1 as one of the nonzero elements — in any case one checks the only possible ring structure compatible with idempotence and distributivity is the product ring $\mathbb{F}_2 \times \mathbb{F}_2$. Concretely, label the four elements $\{0, 1, u, v\}$ with $u + v = 1$, $u^2 = u$, $v^2 = v$; then multiplication must satisfy $uv = 0$ (since $u(1 - u) = 0$), so the multiplication table is exactly that of $\mathbb{F}_2 \times \mathbb{F}_2$. Thus up to isomorphism there is a unique Boolean ring of order 4, namely $\mathbb{F}_2 \times \mathbb{F}_2$.

(One can also argue using the general structure theorem: every finite Boolean ring is isomorphic to a finite product of copies of $\mathbb{F}_2$; for size 4 this forces $\mathbb{F}_2 \times \mathbb{F}_2$.)

7. *Cardinal of a finite Boolean ring is a power of 2.* From (2) the additive group $(R, +)$ satisfies $2a = 0$ for all a , so it is a vector space over $\mathbb{F}_2$. If R is finite and the additive group has dimension n over $\mathbb{F}_2$ then $|R| = 2^n$. Therefore the cardinal of any finite Boolean ring is a power of 2.

Exercise 27 (Euclidean rings). Let R be a ring. We say that R is an *Euclidean ring* if R is an integral domain and there exists a map

$$v : R \setminus \{0\} \longrightarrow \mathbb{N}$$

(the Euclidean valuation) such that for all $a, b \in R$ with $b \neq 0$ there exist $q, r \in R$ satisfying

$$a = qb + r \quad \text{with either } r = 0 \text{ or } v(r) < v(b).$$

Claim. $\mathbb{Z}$ and $K[x]$ (polynomial ring over a field K) are Euclidean rings; moreover $\mathbb{Z}[i]$ (Gaussian integers) is Euclidean with valuation $v(z) = |z|^2$.

Solution.

1. $\mathbb{Z}$ is Euclidean. Take $v(n) = |n|$ for $n \in \mathbb{Z} \setminus \{0\}$. Given $a, b \in \mathbb{Z}$ with $b \neq 0$, the ordinary integer division algorithm yields integers q, r with

$$a = qb + r, \quad 0 \leq r < |b|.$$

If $r = 0$ we are done; otherwise $0 < r < |b|$ so $v(r) = |r| < |b| = v(b)$. Thus $\mathbb{Z}$ is an Euclidean domain (hence an integral domain and a PID).

2. $K[x]$ is Euclidean (for a field K). Let $v(f) = \deg f$ for nonzero $f \in K[x]$ (degree 0). Given $a(x), b(x) \in K[x]$ with $b \neq 0$, the polynomial division algorithm (since K is a field) provides unique $q(x), r(x) \in K[x]$ with

$$a(x) = q(x)b(x) + r(x) \quad \text{and} \quad r(x) = 0 \text{ or } \deg r < \deg b.$$

If $r = 0$ we are done; otherwise $v(r) = \deg r < \deg b = v(b)$. Hence $K[x]$ is Euclidean.

3. $\mathbb{Z}[i]$ is Euclidean with $v(z) = |z|^2$. Let $z = a + bi$ denote a Gaussian integer ($a, b \in \mathbb{Z}$); define the norm $N(z) = |z|^2 = a^2 + b^2 \in \mathbb{N}$. We first note N is multiplicative:

$$N(z_1 z_2) = N(z_1)N(z_2) \quad \text{for } z_1, z_2 \in \mathbb{Z}[i],$$

so $\mathbb{Z}[i]$ is an integral domain and N takes positive integer values on nonzero elements.

To show the Euclidean division property: take $\alpha, \beta \in \mathbb{Z}[i]$ with $\beta \neq 0$. Consider the complex quotient

$$w := \frac{\alpha}{\beta} \in \mathbb{C}.$$

Choose $q \in \mathbb{Z}[i]$ (a Gaussian integer) such that q is a nearest Gaussian integer to w : equivalently write $w = x + iy$ with $x, y \in \mathbb{R}$ and pick integers $m, n \in \mathbb{Z}$ with $|x - m| \leq \frac{1}{2}$, $|y - n| \leq \frac{1}{2}$; set $q = m + ni$. Define the remainder

$$r := \alpha - q\beta \in \mathbb{Z}[i].$$

Then

$$\frac{r}{\beta} = w - q = (x - m) + i(y - n),$$

and by the choice of m, n we have

$$\left| \frac{r}{\beta} \right|^2 = (x - m)^2 + (y - n)^2 \leq \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 = \frac{1}{2}.$$

Multiplying by $N(\beta) = |\beta|^2$ yields

$$N(r) = |r|^2 = |\beta|^2 \cdot |w - q|^2 = N(\beta) \cdot |w - q|^2 \leq \frac{1}{2} N(\beta) < N(\beta).$$

Thus $r = 0$ or $N(r) < N(\beta)$, which is exactly the Euclidean remainder property for the valuation $v(\cdot) = N(\cdot) = |\cdot|^2$.

Therefore for every $\alpha, \beta \in \mathbb{Z}[i]$ with $\beta \neq 0$ there exist $q, r \in \mathbb{Z}[i]$ with $\alpha = q\beta + r$ and either $r = 0$ or $v(r) < v(\beta)$. Hence $\mathbb{Z}[i]$ is an Euclidean domain.

Conclusion. $\mathbb{Z}$, $K[x]$ (for any field K), and $\mathbb{Z}[i]$ (with $v(z) = |z|^2$) are Euclidean rings. Consequently each is an integral domain in which the Euclidean algorithm (and hence gcd computations and Bézout relations) can be carried out.